

INVERTER GEN5

OPEN INVERTER PLATFORM

Hardware Release Report

Chassis	Chassis2
Generated	2026-07-16
BOM lines	106 (98 priced)

Cost at quantity

Units	Estimated total
1	\$2,403.52
2	\$4,739.58
3	\$7,070.09
5	\$11,744.59
10	\$23,433.45

Order Summary

In-House Assembly

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-WH-TSBCM-A	CapacitorTempSenseHarness	-	N/A	N/A
4	HW-C2-WH-CS-A	CurrentSenseHarness	-	N/A	N/A
1	HW-C2-WH-GDVS-A	GDVSHarness	-	N/A	N/A
2	HW-C2-WH-TSIGBT-A	TempSenseHarness	-	N/A	N/A
				Subtotal	\$0.00

Digi-Key

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-HW-3M156058-ND -A	3M156058-ND	3M156058-ND	\$46.9800	\$46.98
				Subtotal	\$46.98

McMaster-Carr

Qty	Internal P/N	Description	Part Number	Unit	Total
8 (1pk)	HW-C2-MECH-90128A232 -A	90128A232	90128A232	\$10.5500/pk	\$10.55
5 (1pk)	HW-C2-MECH-90225A102 -A	90225A102	90225A102	\$12.8000/pk	\$12.80
18 (1pk)	HW-C2-MECH-91292A134 -A	91292A134	91292A134	\$11.7200/pk	\$11.72
4 (1pk)	HW-C2-MECH-91294A220 -A	91294A220	91294A220	\$12.7400/pk	\$12.74
6 (1pk)	HW-C2-MECH-91502A180 -A	91502A180	91502A180	\$19.6500/pk	\$19.65
6 (1pk)	HW-C2-MECH-91502A183 -A	91502A183	91502A183	\$17.2700/pk	\$17.27
6	HW-C2-MECH-94669A190 -A	94669A190	94669A190	\$1.3900	\$8.34
6	HW-C2-MECH-94669A196 -A	94669A196	94669A196	\$2.9600	\$17.76
6	HW-C2-MECH-94669A199 -A	94669A199	94669A199	\$3.1000	\$18.60
6	HW-C2-MECH-94669A202 -A	94669A202	94669A202	\$3.2200	\$19.32
25 (1pk)	HW-C2-MECH-98044A224 -A	98044A224	98044A224	\$14.8800/pk	\$14.88
				Subtotal	\$163.63

Mouser

Qty	Internal P/N	Description	Part Number	Unit	Total
8	HW-C2-HW-0022011042-A	0022011042	-	N/A	N/A
32	HW-C2-HW-08-50-0113-A	08-50-0113	-	N/A	N/A
70	HW-C2-CAP-100N-A	100nF 100V	80-C1210C104K1RAC	\$0.1610	\$11.27
27	HW-C2-RES-10K-A	10k 1210	603-AC1210JR-0710K L	\$0.0500	\$1.35
21	HW-C2-CAP-10N-A	10nF 500V	187-CL32B103KGFNN NE	\$0.1430	\$3.00
37	HW-C2-CAP-10U-A	10uF 25V	187-CL32A106KAULN NE	\$0.1340	\$4.96
4	HW-C2-RES-120-A	120 1210	603-AC1210FR-07120 RL	\$0.1000	\$0.40
17	HW-C2-RES-1K-A	1k 1210	603-RC1210FR-071KL	\$0.0490	\$0.83
8	HW-C2-CAP-1U-A	1uf 50V	187-CL32B105KBHNN NE	\$0.2400	\$1.92
2	HW-C2-CAP-2U2-A	2.2uf 1210 100v	80-C1210C225J1RAU TO	\$0.5200	\$1.04
12	HW-C2-RES-2R7-A	2.7 ohm 1210	667-ERJ-14BQF2R7U	\$0.2340	\$2.81
16	HW-C2-RES-20K-A	20k 1210 resistor	667-ERJ-P14F2002U	\$0.2450	\$3.92
4	HW-C2-HW-22-04-1041-A	22-04-1041	538-22-04-1041	\$0.8900	\$3.56
1	HW-C2-RES-220-A	220 1210	603-RC1210FR-07220 RL	\$0.1000	\$0.10
6	HW-C2-CAP-220P-A	220pF 100V C0G 1210	80-C1210C221J2G	\$0.8800	\$5.28
4	HW-C2-CAP-22P-A	22pF 50V	80-C1210C220J5G	\$0.7400	\$2.96
3	HW-C2-RES-3K-A	3k 1210 1%	603-RC1210FR-073KL	\$0.1000	\$0.30
3	HW-C2-RES-4K7-A	4.7k 1210	71-CRCW12104K70F KEAH	\$0.5900	\$1.77
3	HW-C2-HW-455-2266-ND-A	455-2266-ND	306-XHP-2	\$0.1000	\$0.30
25	HW-C2-HW-464762-E-A	464762-E	305-464762	\$0.3970	\$9.93
1	HW-C2-RES-499K-A	499kohm 1210 resistor	667-ERJ-P14F4993U	\$0.4400	\$0.44
5	HW-C2-HW-524265-E-A	524265-E	305-524265-E	\$7.8700	\$39.35
5	HW-C2-HW-544959-E-A	544959-E	-	N/A	N/A
2	HW-C2-RES-560-A	560ohm 1210 resistor	667-ERJ-P14F5600U	\$0.4400	\$0.88
1	HW-C2-HW-776231-1-A	776231-1	571-776231-1	\$16.6900	\$16.69
2	HW-C2-IC-8MHZ-A	8mhz hc49-4h crystal	710-830003156B	\$0.5000	\$1.00
8	HW-C2-HW-971150581-A	971150581	710-971150581	\$0.6000	\$4.80
8	HW-C2-HW-971250581-A	971250581	710-971250581	\$1.1000	\$8.80
8	HW-C2-HW-971350581-A	971350581	710-971350581	\$1.4800	\$11.84
8	HW-C2-HW-9774030592R-A	9774030592R	710-9774030592R	\$1.5700	\$12.56

Qty	Internal P/N	Description	Part Number	Unit	Total
2	HW-C2-HW-ACT45B-101-2P-TL003-A	ACT45B-101-2P-TL003	810-ACT45B1012PTL003	\$1.5300	\$3.06
2	HW-C2-HW-AWHS105D00G-A	AWHS105D00G	523-AWHS105D00G	\$1.1100	\$2.22
3	HW-C2-HW-B2B-XH-AM_LF__SN_-A	B2B-XH-AM_LF__SN_	306-B2B-XH-AMLFSP	\$0.1000	\$0.30
12	HW-C2-HW-B58031U9254M062-A	B58031U9254M062	871-B58031U9254M062	\$4.6500	\$55.80
12	HW-C2-HW-BAT54WS-7-F-A	BAT54WS-7-F	621-BAT54WS-F	\$0.1240	\$1.49
5	HW-C2-HW-BTS462TATMA1A-A	BTS462TATMA1	726-BTS462TATMA1	\$2.6600	\$13.30
3	HW-C2-HW-CM600DY-24T-A	CM600DY-24T	CM600DY-24T	\$323.5000	\$970.50
1	HW-C2-IC-CY15B102Q-SXET-A	CY15B102Q-SXET	727-CY15B102Q-SXET	\$18.1400	\$18.14
1	HW-C2-HW-DRDNB21D-7-A	DRDNB21D-7	621-DRDNB21D-7	\$0.5600	\$0.56
1	HW-C2-HW-EC7BW-110S12-A	EC7BW-110S12	418-EC7BW-110S12	\$48.8600	\$48.86
12	HW-C2-HW-FDFNYD1-110-5-A	FDFNYD1-110(5)	161-FDFNYD1-1105	\$0.4800	\$5.76
4	HW-C2-IC-ISO1042BDWVR-A	ISO1042BDWVR	595-ISO1042BDWVR	\$4.4400	\$17.76
1	HW-C2-IC-ISO1212DBQ-A	ISO1212DBQ	595-ISO1212DBQ	\$1.7300	\$1.73
4	HW-C2-HW-LA37S600S05KM-A	LA37S600S05KM	838-LA37S600S05KM	\$19.9600	\$79.84
1	HW-C2-HW-LT3080IQ-PBF-A	LT3080IQ#PBF	584-LT3080IQ#PBF	\$8.5400	\$8.54
2	HW-C2-HW-LTST-C230CKT-A	LTST-C230CKT	859-LTST-C230CKT	\$0.4100	\$0.82
2	HW-C2-HW-LTST-C230GKT-A	LTST-C230GKT	859-LTST-C230GKT	\$0.3600	\$0.72
1	HW-C2-IC-MAX22530AWE-A	MAX22530AWE+	700-MAX22530AWE+	\$15.6000	\$15.60
1	HW-C2-HW-MF-RHT200_32-2-A	MF-RHT200_32-2	652-MF-RHT200/32-2	\$0.6100	\$0.61
6	HW-C2-HW-MGJ2D121509MPC-R7-A	MGJ2D121509MPC-R7	580-MGJ2D121509MPC-R7	\$6.7700	\$40.62
6	HW-C2-HW-MKP1848S61010JY5B-A	MKP1848S61010JY5B	75-MKP1848S61010JY5B	\$7.5400	\$45.24
5	HW-C2-HW-MM3Z3V0ST1G-A	MM3Z3V0ST1G	863-MM3Z3V0ST1G	\$0.1700	\$0.85
6	HW-C2-IC-NCV57100DWR2G-A	NCV57100DWR2G	863-NCV57100DWR2G	\$2.9800	\$17.88
1	HW-C2-HW-NXFT15XH103FEAB021-A	NXFT15XH103FEAB021	81-NXFT15XH103FEAB021	\$0.3400	\$0.34

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-HW-RD7-12S033R-A	RD7-12S033R	737-RD7-12S033R	\$6.7700	\$6.77
3	HW-C2-HW-RKE-1205S_H-A	RKE-1205S_H	919-RKE-1205S/H	\$5.1700	\$15.51
2	HW-C2-HW-RTS103C1R2M6L201-A	RTS103C1R2M6L201	527-RTS103C1R2M6L201	\$2.2700	\$4.54
2	HW-C2-HW-S3N-A	S3N	512-S3N	\$0.6700	\$1.34
2	HW-C2-HW-SSW-103-01-L-D-A	SSW-103-01-L-D	200-SSW10301LD	\$1.3400	\$2.68
1	HW-C2-HW-SSW-104-01-L-D-A	SSW-104-01-L-D	200-SSW10401LD	\$1.6900	\$1.69
2	HW-C2-HW-SSW-108-01-L-D-A	SSW-108-01-L-D	200-SSW10801LD	\$2.3500	\$4.70
1	HW-C2-CONN-SSW-110-01-F-D-A	SSW-110-01-F-D	200-SSW11001FD	\$2.4100	\$2.41
1	HW-C2-HW-SSW-110-01-F-D-A	SSW-110-01-F-D	-	N/A	N/A
1	HW-C2-IC-STM32G474RCT3-A	STM32G474RCT3	511-STM32G474RCT3	\$8.3100	\$8.31
1	HW-C2-IC-STM32H723ZGT6-A	STM32H723ZGT6	511-STM32H723ZGT6	\$13.2500	\$13.25
1	HW-C2-HW-STS41A-AWLB-R3-A	STS41A-AWLB-R3	403-STS41A-AWLB-R3	\$1.1300	\$1.13
6	HW-C2-HW-STTH112A-A	STTH112A	511-STTH112A	\$0.5900	\$3.54
6	HW-C2-HW-SXH-001T-P0.6-A	SXH-001T-P0.6	306-SXH-001T-P0.6	\$0.1000	\$0.60
4	HW-C2-HW-SZMMSZ4678T1G-A	SZMMSZ4678T1G	863-SZMMSZ4678T1G	\$0.4300	\$1.72
21	HW-C2-HW-TESTPOINT-TH-RED-A	Testpoint TH, RED	534-5000	\$0.2930	\$6.15
16	HW-C2-RES-250K-A	Thick Film Resistors - SMD 1/2watt 249Kohms 1%	71-CRCW1210-249K-E3	\$0.0700	\$1.12
1	HW-C2-HW-TPS389006A-DJRTER-A	TPS389006ADJRTER	595-TPS389006ADJRTER	\$5.1000	\$5.10
2	HW-C2-HW-TS40-66-50-R-260-SMT-TR-A	TS40-66-50-R-260-SMT-TR	179-TS406650R260S	\$0.4800	\$0.96
2	HW-C2-HW-TSW-103-18-G-D-A	TSW-103-18-G-D	200-TSW10318GD	\$1.0100	\$2.02
2	HW-C2-HW-TSW-108-18-F-D-A	TSW-108-18-F-D	200-TSW10818FD	\$1.4500	\$2.90
1	HW-C2-HW-TSW-110-18-T-D-A	TSW-110-18-T-D	200-TSW11018TD	\$1.5800	\$1.58
1	HW-C2-CONN-TSW-110-21-L-D-LL-A	TSW-110-21-L-D-LL	200-TSW11021LDLL	\$2.2700	\$2.27
13	HW-C2-HW-UCM1H101MCL1GS-A	UCM1H101MCL1GS	647-UCM1H101MCL1GS	\$0.3250	\$4.23
60	HW-C2-HW-UCS2D331MHD-A	UCS2D331MHD	647-UCS2D331MHD	\$2.1600	\$129.60

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-HW-UJ2-B-HR-G-V-A	UJ2-B-HR-G-V	179-UJ2-B-HR-G-V	\$1.7000	\$1.70
18	HW-C2-HW-VY1222M47Y5UQ6TV0-A	VY1222M47Y5UQ6TV0	72-VY1222M47Y5UQ6TV0	\$0.2710	\$4.88
				Subtotal	\$1,733.29

PCB Fabrication

Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-PCB-CTRL-A	ControlBoard	ControlBoard	\$25.2400	\$25.24
1	HW-C2-PCB-DCBUSCAP-A	DCBusCapacitorBoard	DCBusCapacitorBoard	\$7.5400	\$7.54
1	HW-C2-PCB-DCFLT-A	DCBusFilter	DCBusFilter	\$8.4400	\$8.44
1	HW-C2-PCB-GD-A	GateDriver	GateDriver	\$11.6600	\$11.66
1	HW-C2-PCB-IO-A	IOBoard	IOBoard	\$11.6600	\$11.66
				Subtotal	\$64.54

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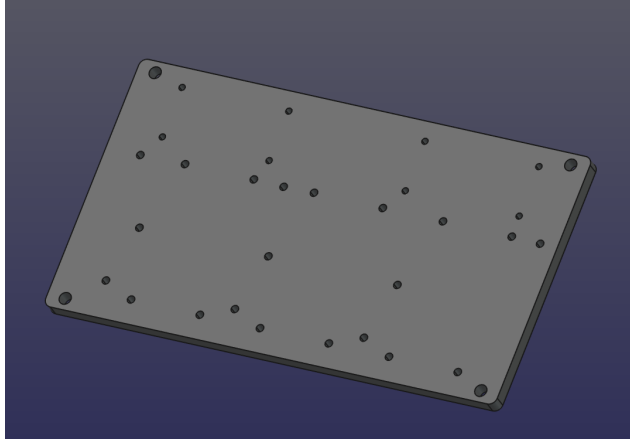
Qty	Internal P/N	Description	Part Number	Unit	Total
1	HW-C2-PLT-BSP-A	HW-C2-BSP-A	HW-C2-BSP-A	\$201.6100	\$201.61
1	HW-C2-PLT-CHSP-A	HW-C2-CHSP-A	HW-C2-CHSP-A	\$33.3100	\$33.31
2	HW-C2-BB-DCLBB-A	HW-C2-DCLBB-A	HW-C2-DCLBB-A	\$50.7100	\$101.42
3	HW-C2-BB-PBB-A	HW-C2-PBB-A	HW-C2-PBB-A	\$19.5800	\$58.74
				Subtotal	\$395.08

Fabrication Package

Item	Kind	Qty	Rev	Price	Ready
ControlBoard	PCB	1	A	\$25.24	yes
DCBusCapacitorBoard	PCB	1	A	\$7.54	yes
DCBusFilter	PCB	1	A	\$8.44	yes
GateDriver	PCB	1	A	\$11.66	yes
IOBoard	PCB	1	A	\$11.66	yes
HW-C2-BSP-A	fab part	1	A	\$201.61	yes
HW-C2-CHSP-A	fab part	1	A	\$33.31	yes
HW-C2-DCLBB-A	fab part	2	A	\$50.71	yes
HW-C2-PBB-A	fab part	3	A	\$19.58	yes
CapacitorTempSenseHarness	harness	-	-	-	yes
CurrentSenseHarness	harness	-	-	-	yes
GDVSHarness	harness	-	-	-	yes
TempSenseHarness	harness	-	-	-	yes

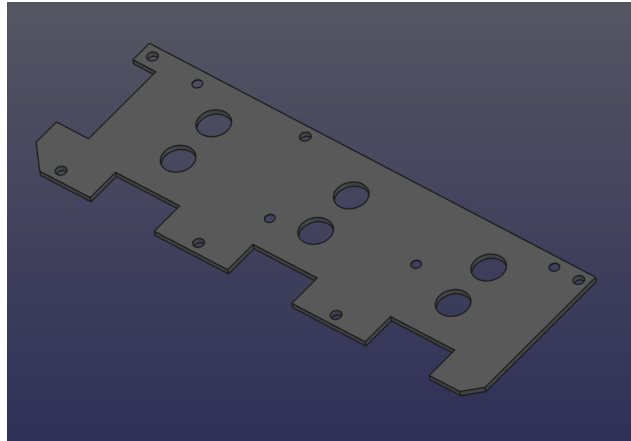
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6061 T6 Aluminum / 9.52 mm / Sheet Cutting / Deburring, Tapping



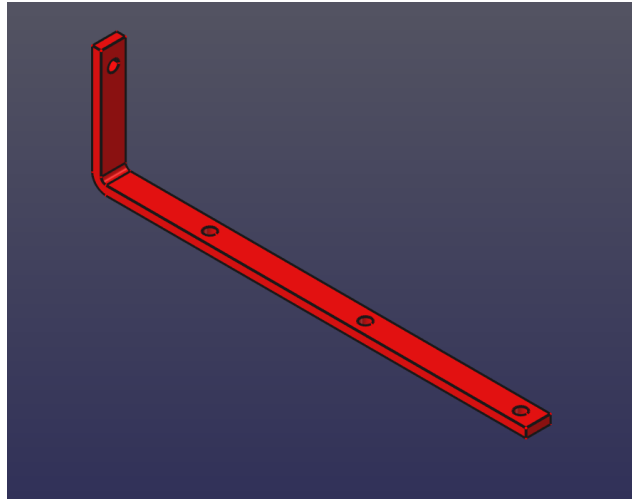
HW-C2-CHSP-A

6061 T6 Aluminum / 3.17 mm / Sheet Cutting / Deburring



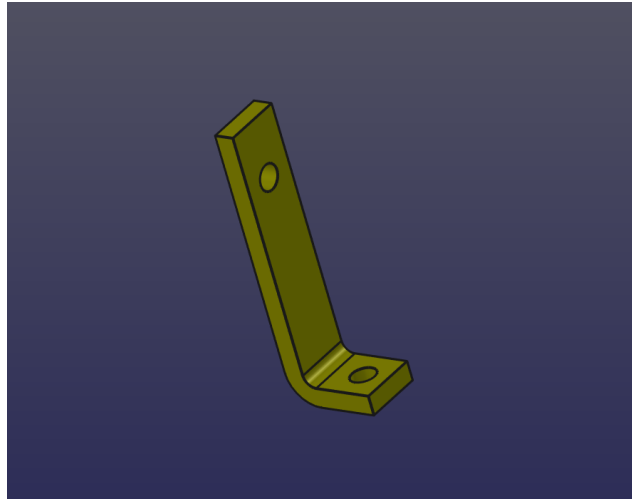
HW-C2-DCLBB-A

Copper / 4.75 mm / Sheet Cutting / Bending



HW-C2-PBB-A

Copper / 4.75 mm / Sheet Cutting / Bending

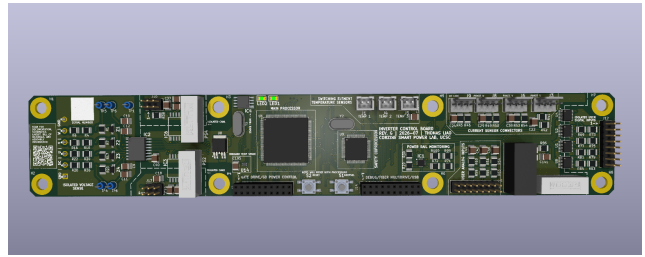
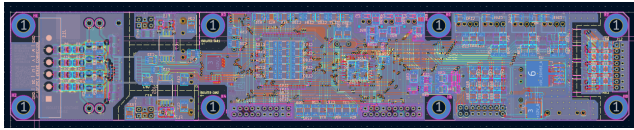


BOMS

printed circuit board — schematic and PCB layers

CONTROLBOARD

HW-C2-PCB-CTRL-A · printed circuit board — schematic and PCB layers



Connector from gate drivers

Includes multiple redundant SSO (six-switch-off) pathways:

- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2

(Gate driver power enable implements 1002, with monitoring feedback).

feedback pins are 0-3.3v binary (3.3v expected when +12v present)

Connector to IO board

Digital Outputs provided here to IOBoard.

SSW-108-01-L-D

J7

SSW-108-01-L-D

J7

SSW-108-01-L-D

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SSW-108-01-L-D

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SSW-108-01-L-D

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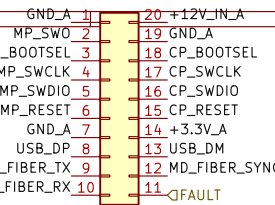
SSW-108-01-L-D

J7

SSW-108-01-L-D

J7

SSW-108-01-L-D



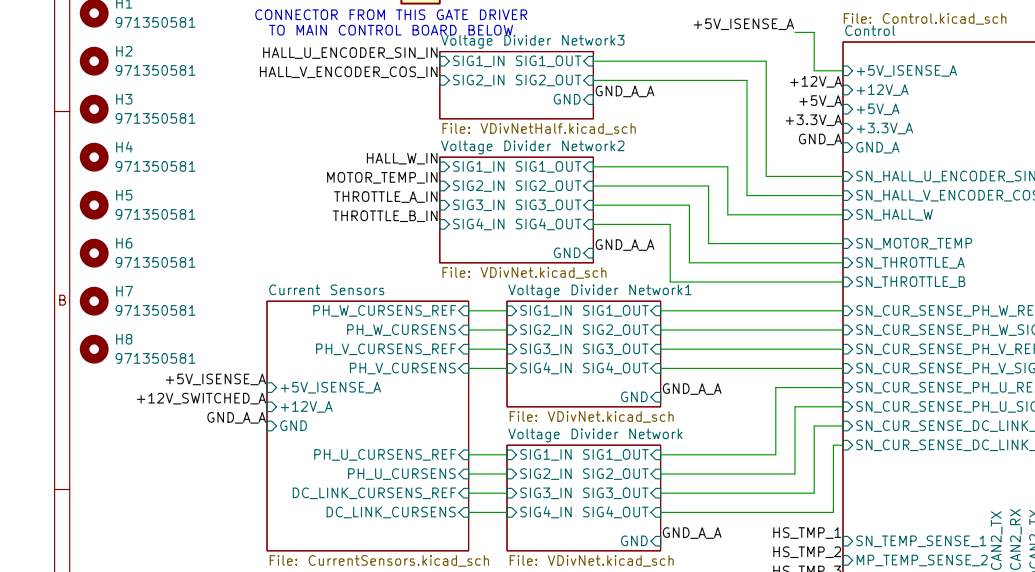
This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)

Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.

Temperature Sensors

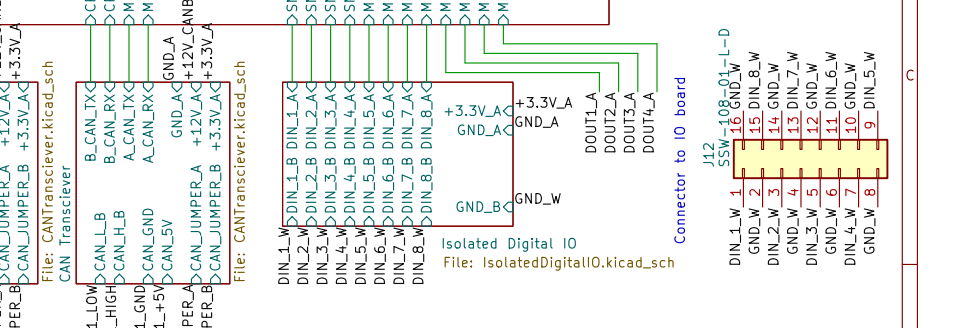
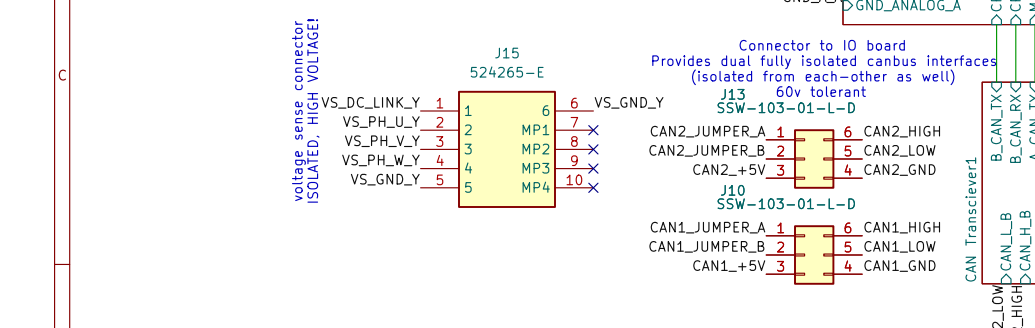


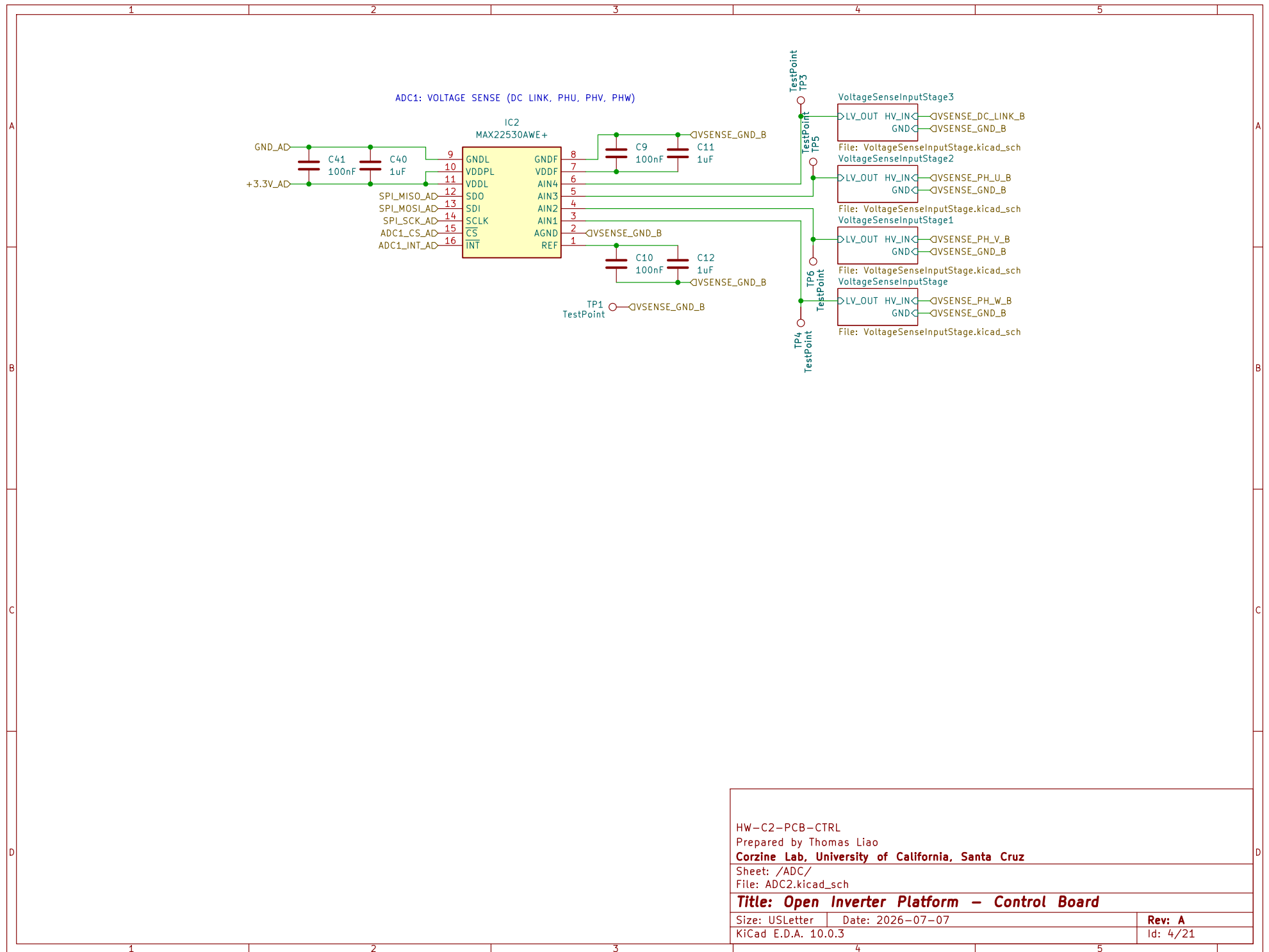
File: TemperatureSensors.kicad_sch



Control system has dual independent mcus, targeting ASIL decomposition. Independently monitored switching commands, current responses, canbus commands + faults

Please see HARA under documentation for additional informaton.





HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/

File: ADC2.kicad_sch

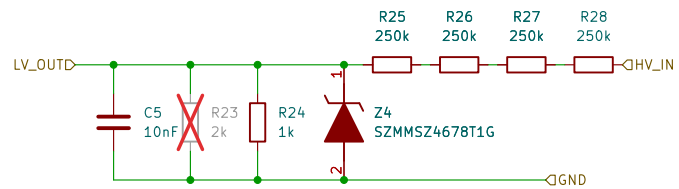
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Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 4/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

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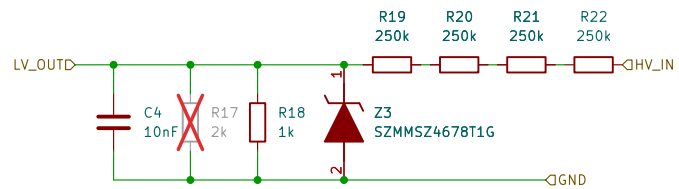
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Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 2/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

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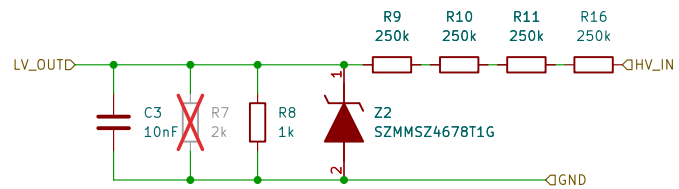
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Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 3/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad_sch

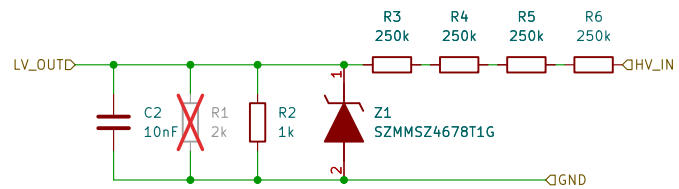
Title: Open Inverter Platform – Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 5/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/

File: VoltageSenseInputStage.kicad_sch

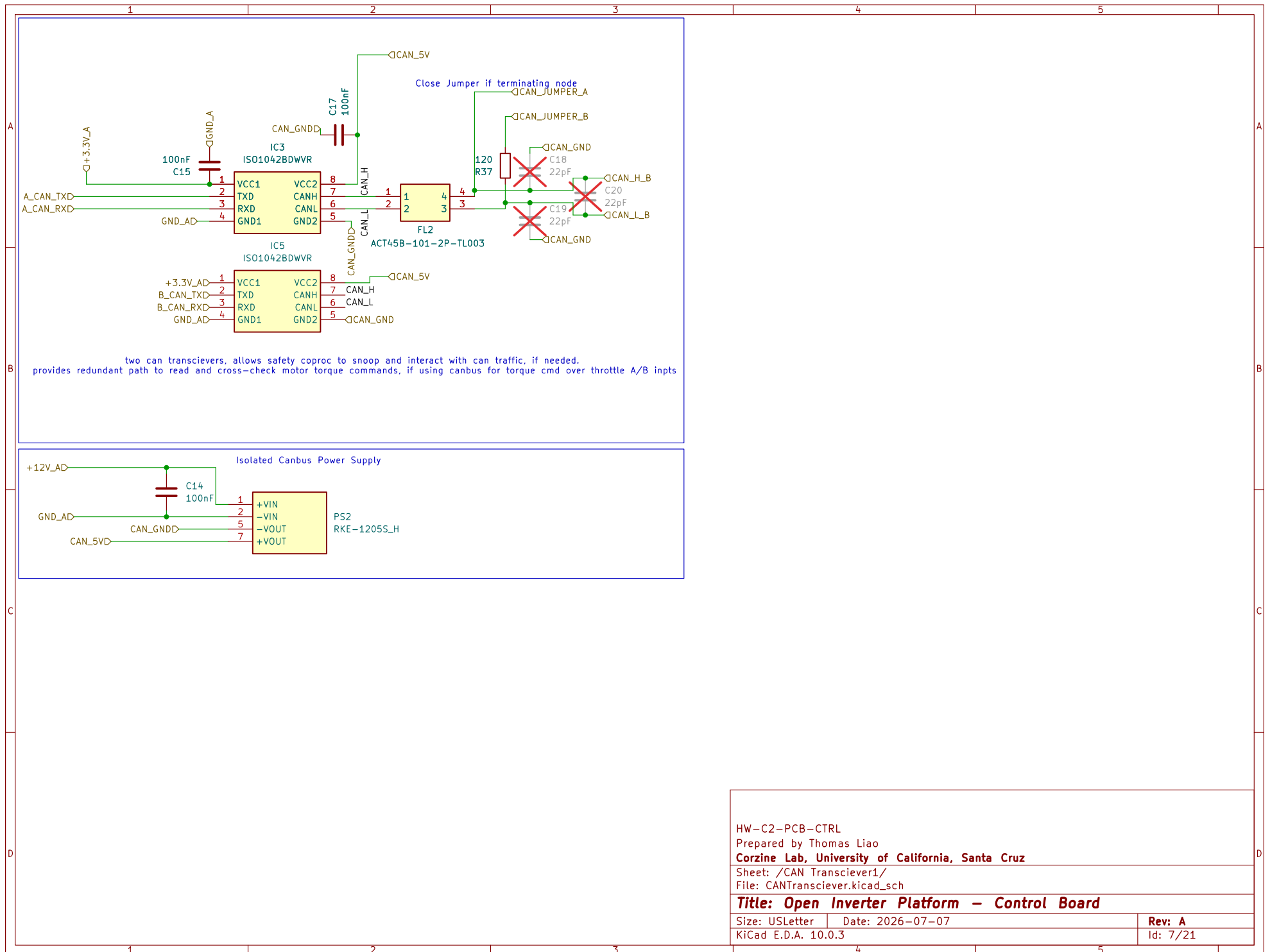
Title: Open Inverter Platform – Control Board

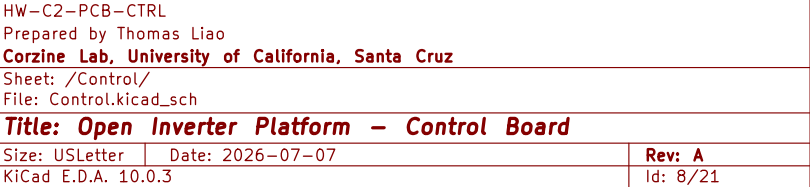
Size: USLetter Date: 2026-07-07

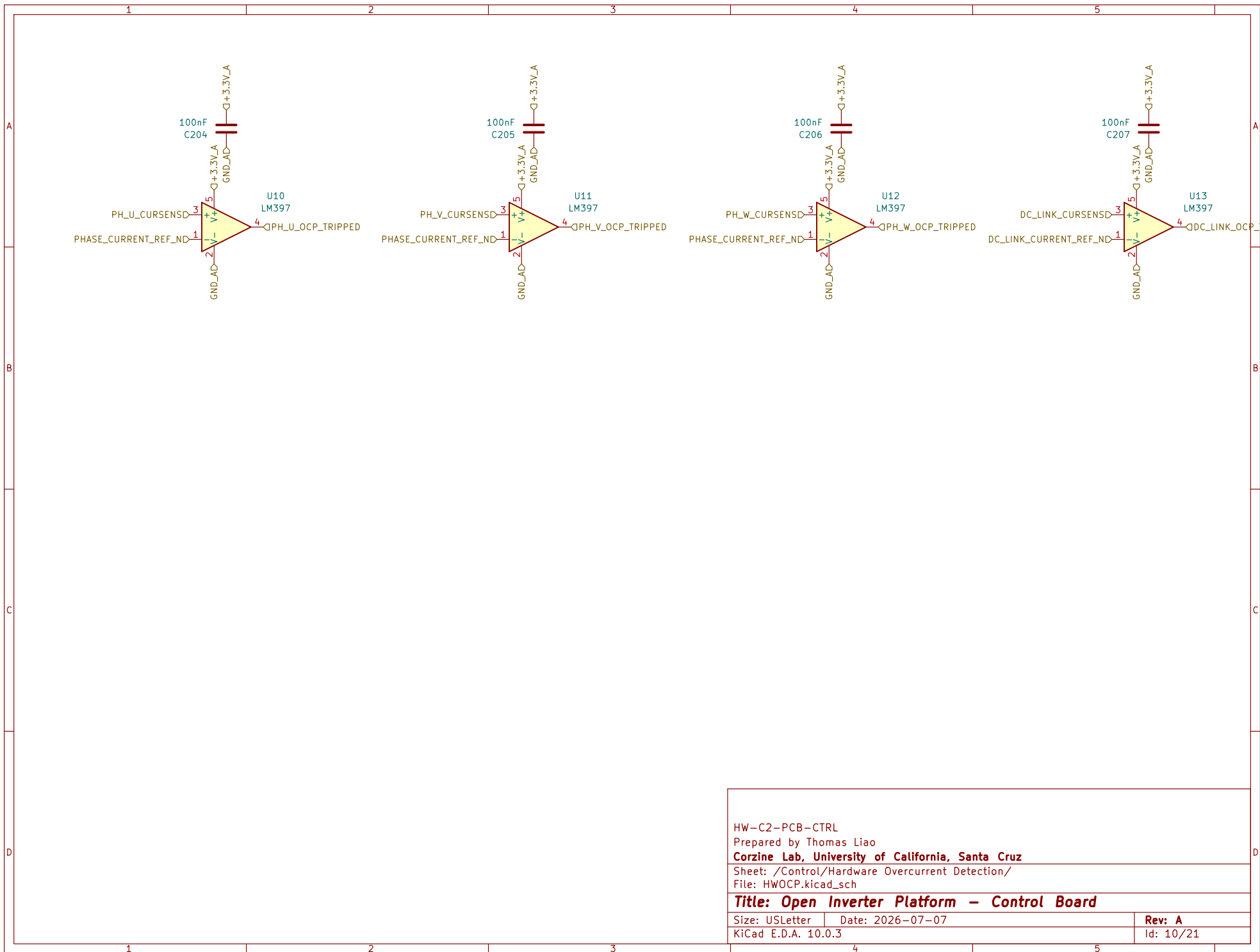
KiCad E.D.A. 10.0.3

Rev: A

Id: 6/21







HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Hardware Overcurrent Detection/

File: HWOCP.kicad_sch

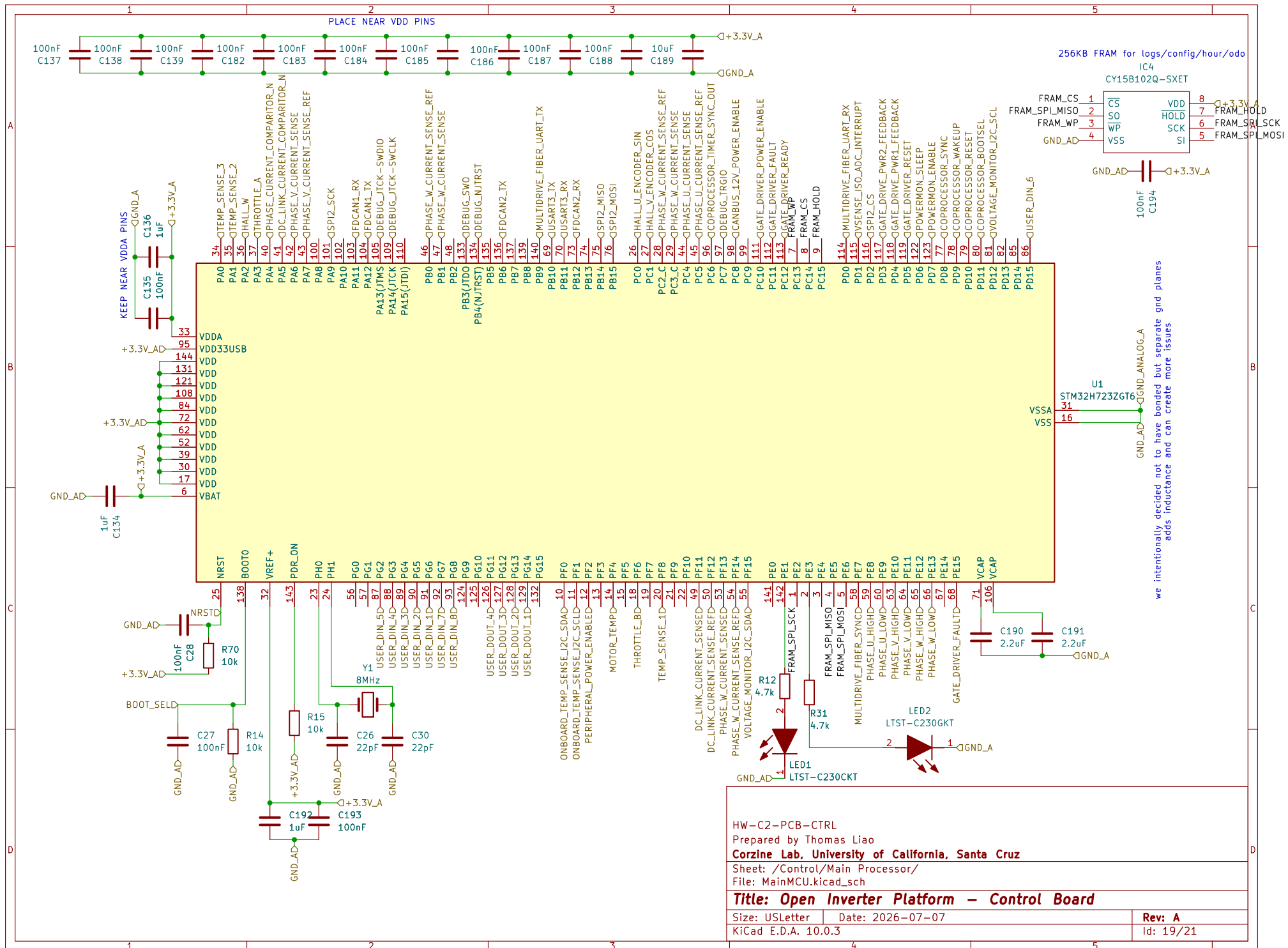
Title: Open Inverter Platform – Control Board

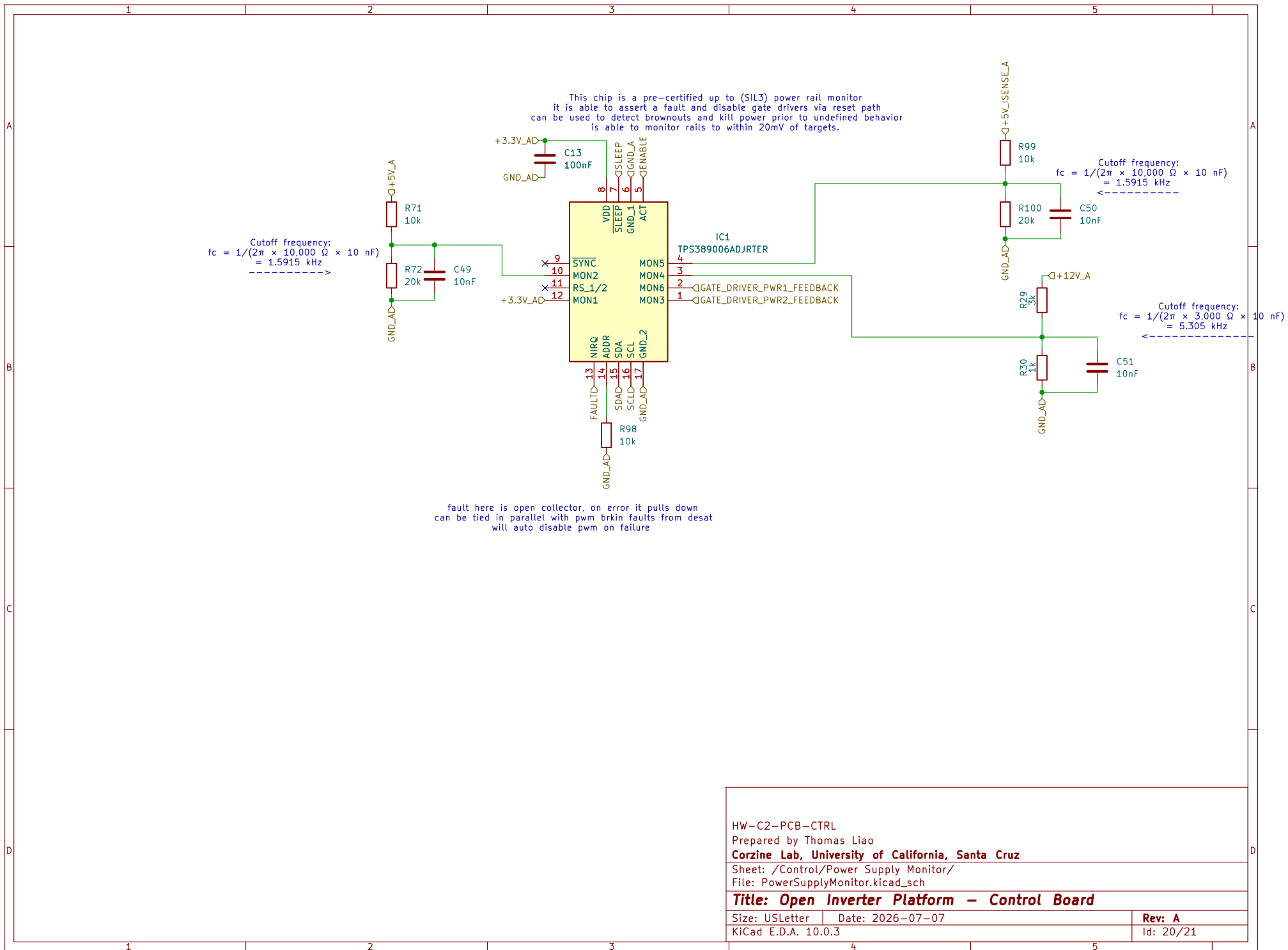
Size: USLetter Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 10/21





HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Power Supply Monitor/

File: PowerSupplyMonitor.kicad_sch

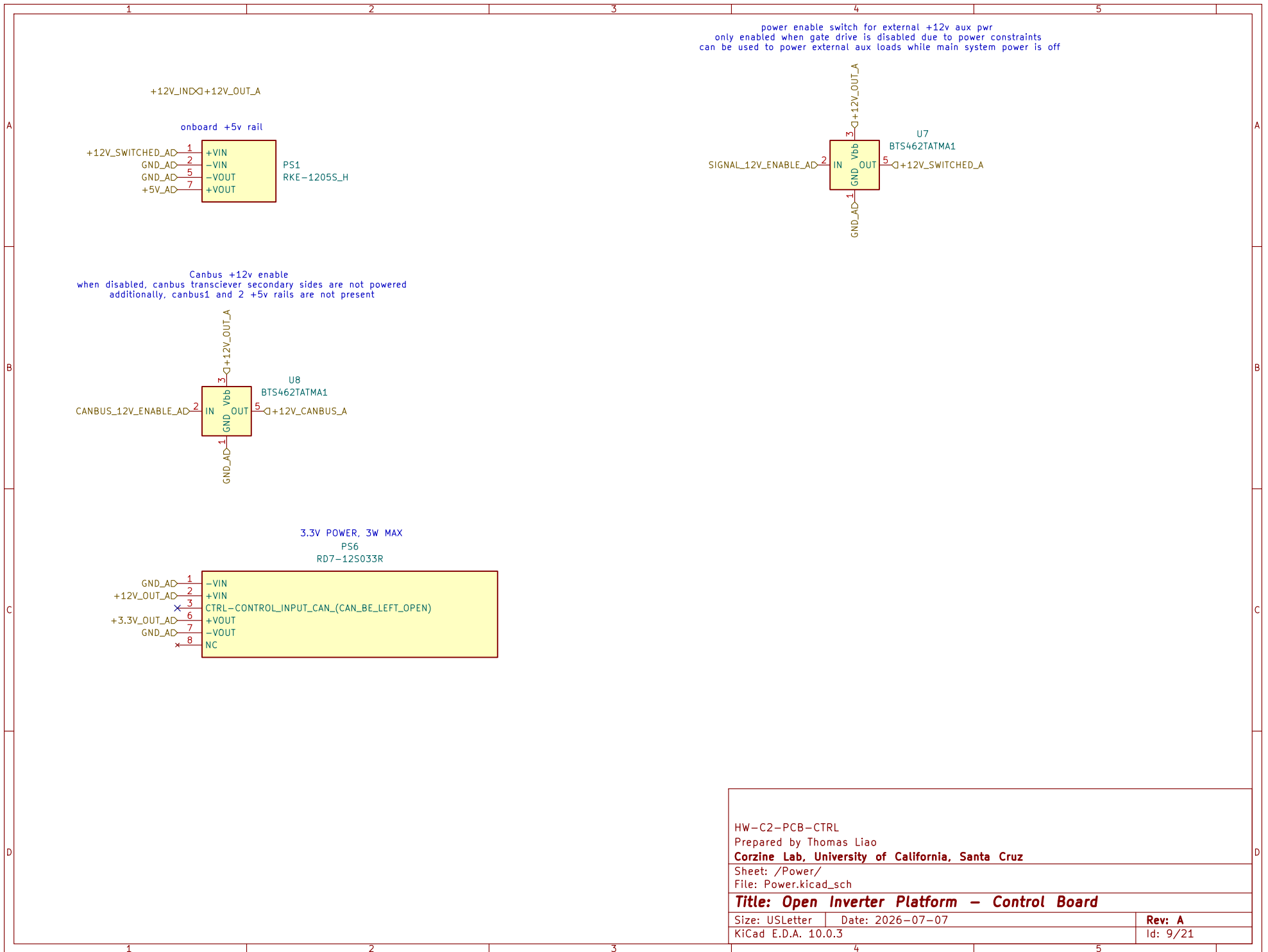
Title: Open Inverter Platform - Control Board

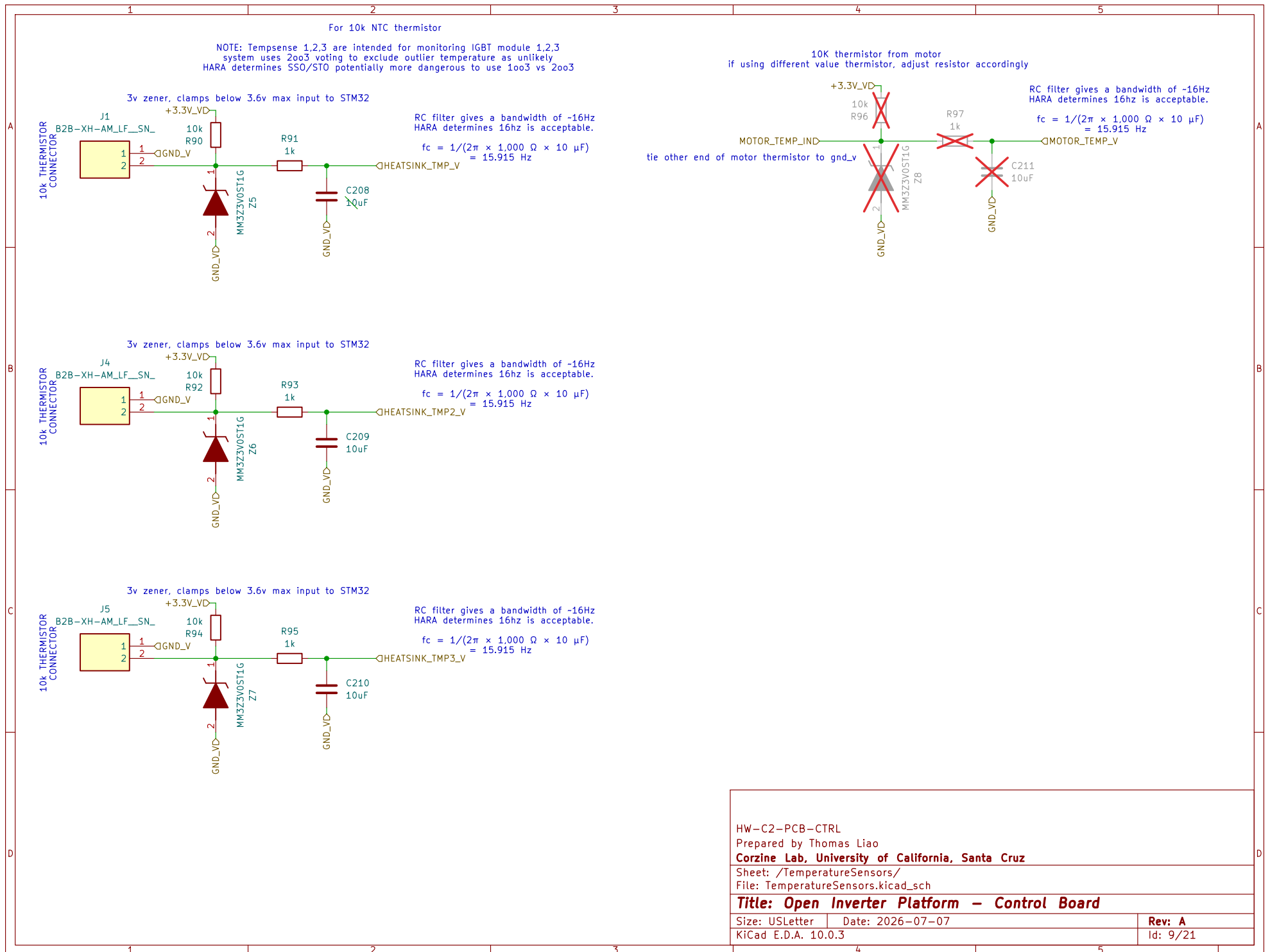
Size: USLetter Date: 2026-07-07

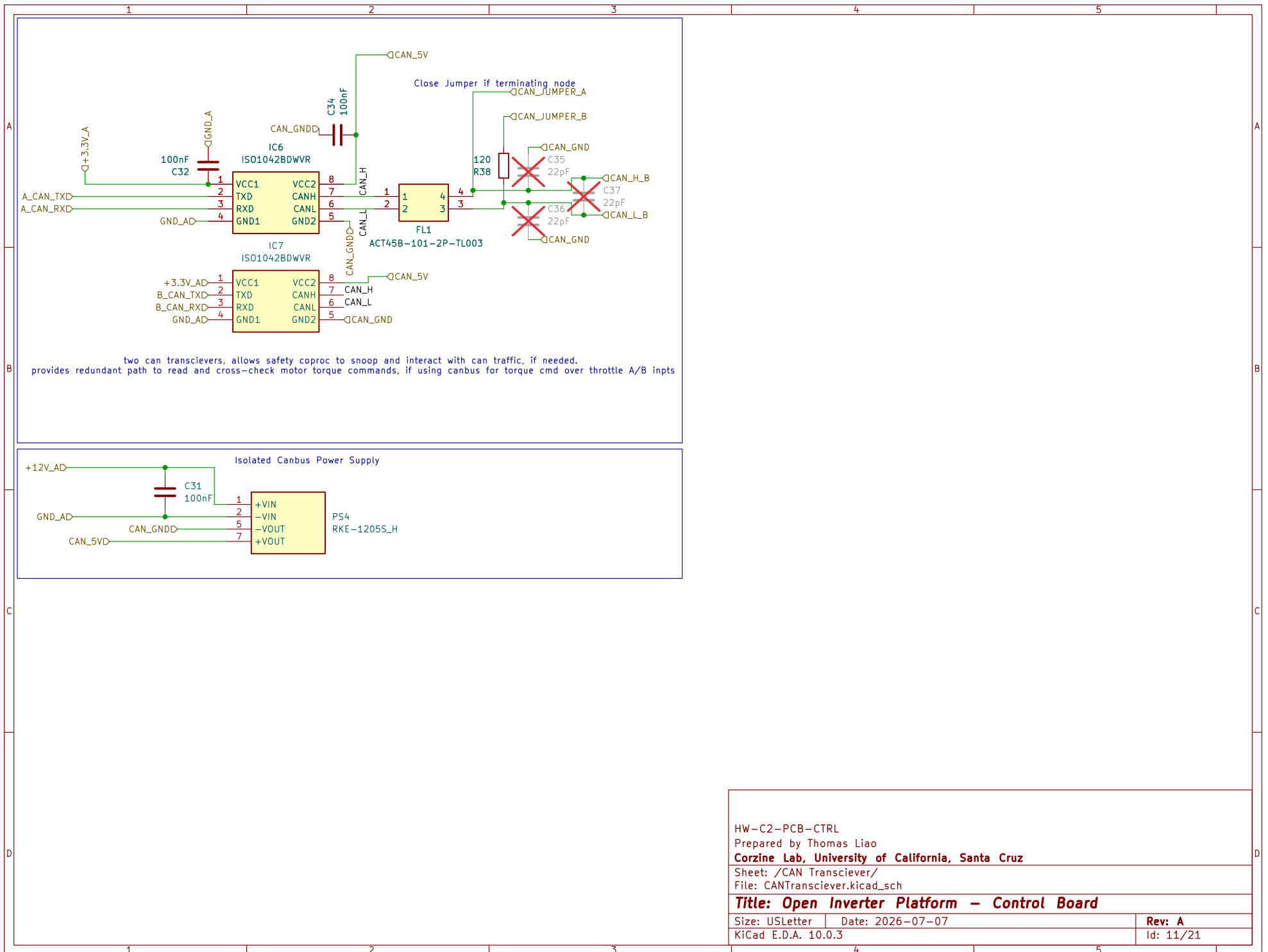
KiCad E.D.A. 10.0.3

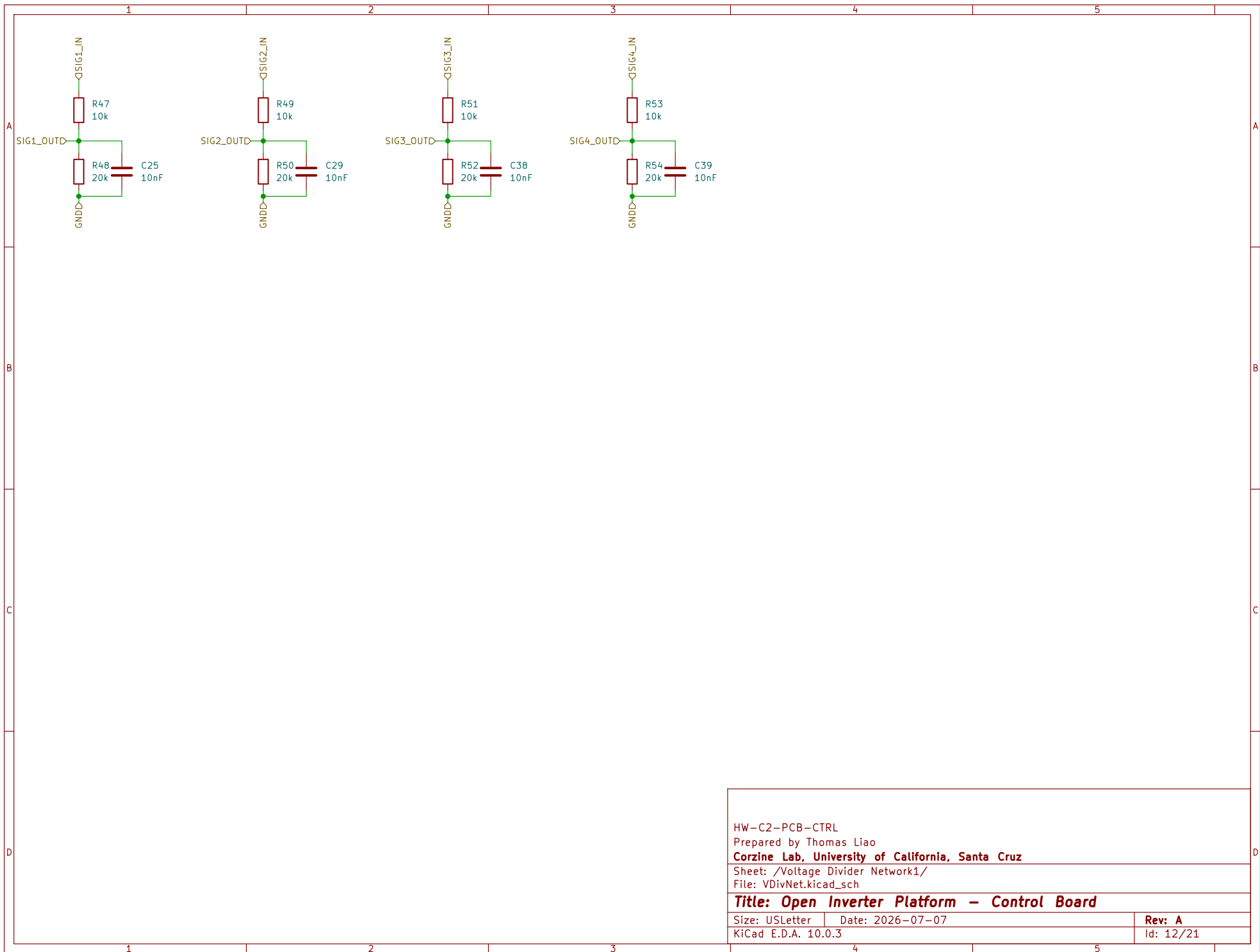
Rev: A

Id: 20/21









HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/

File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

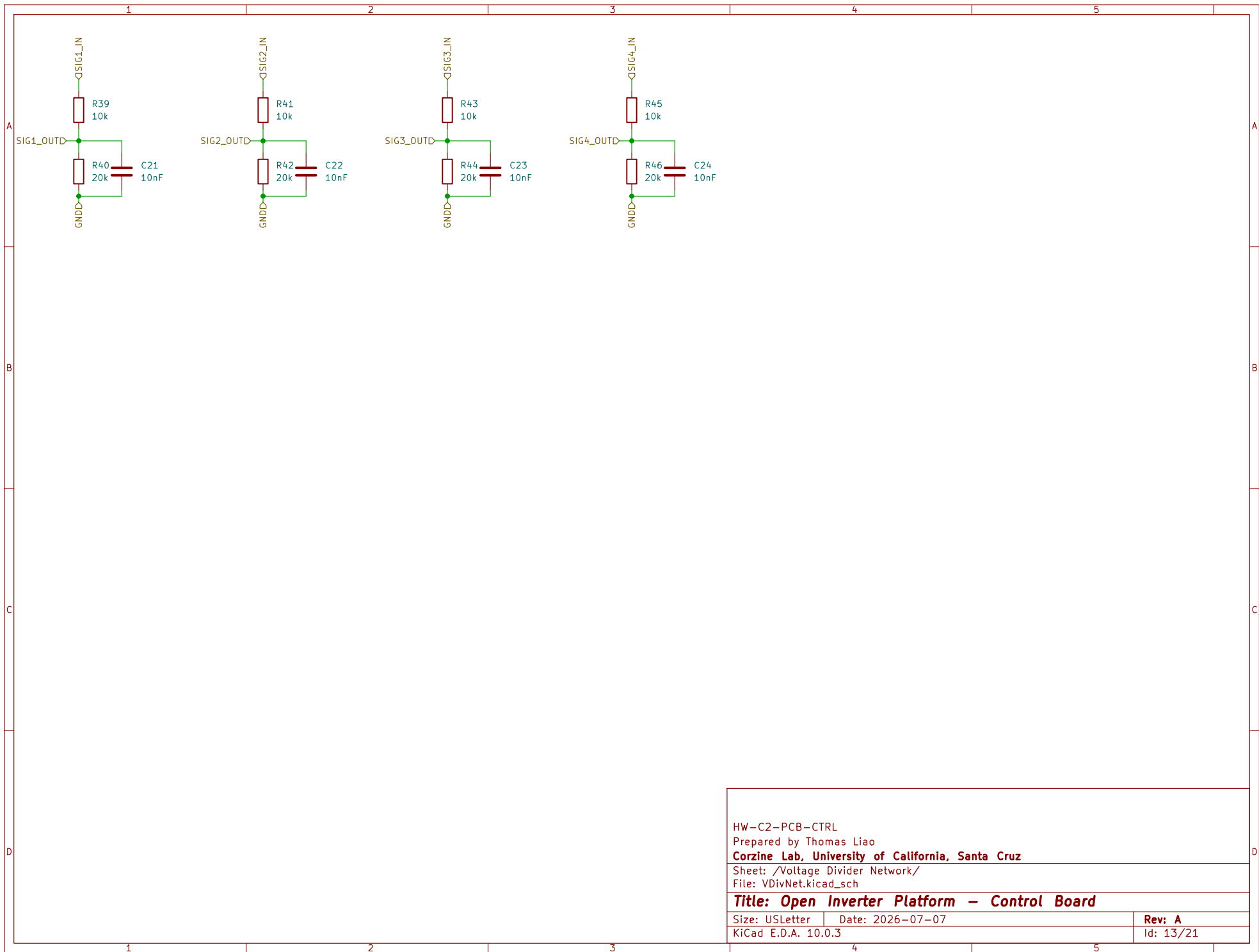
Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 12/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network/

File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

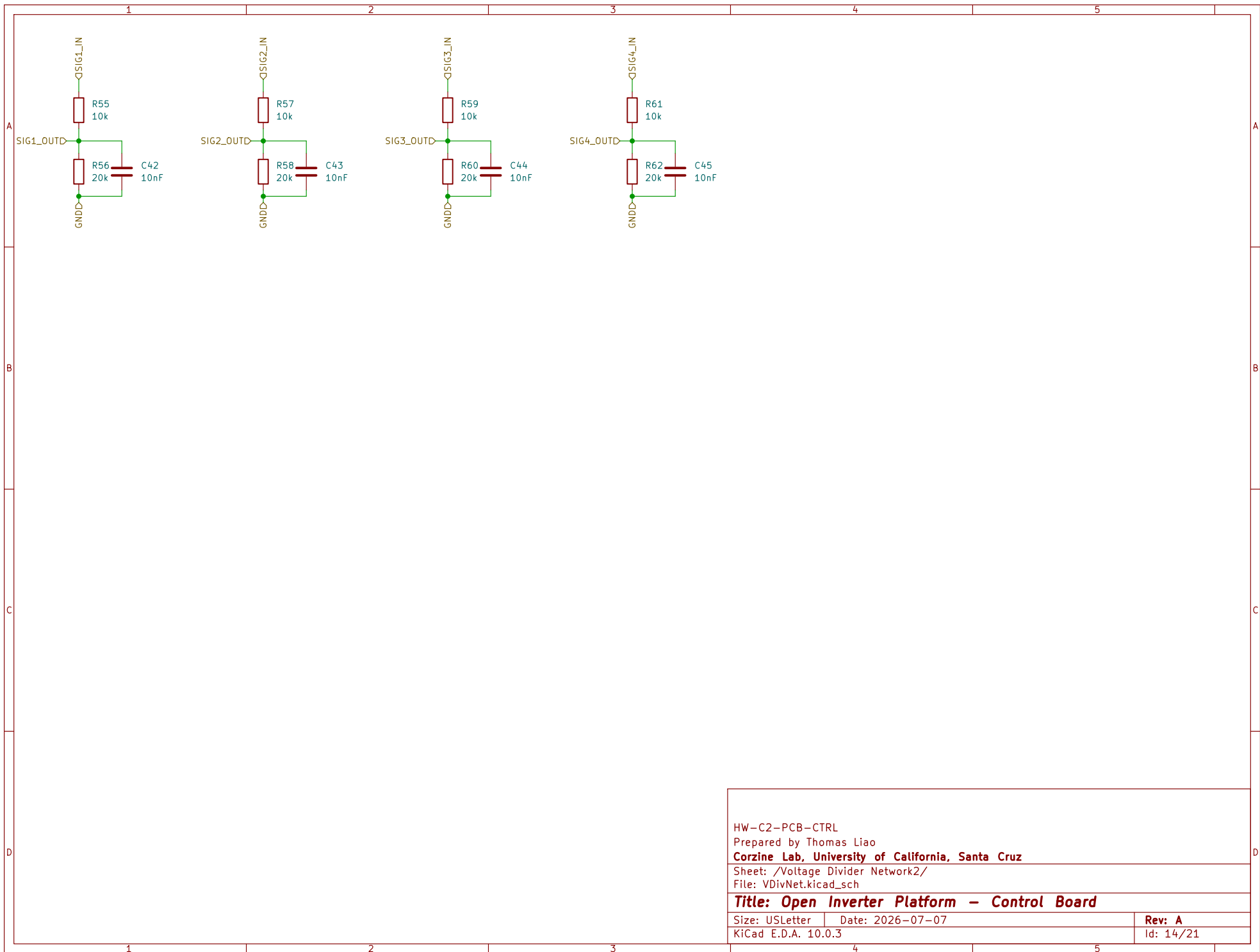
Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 13/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network2/

File: VDivNet.kicad_sch

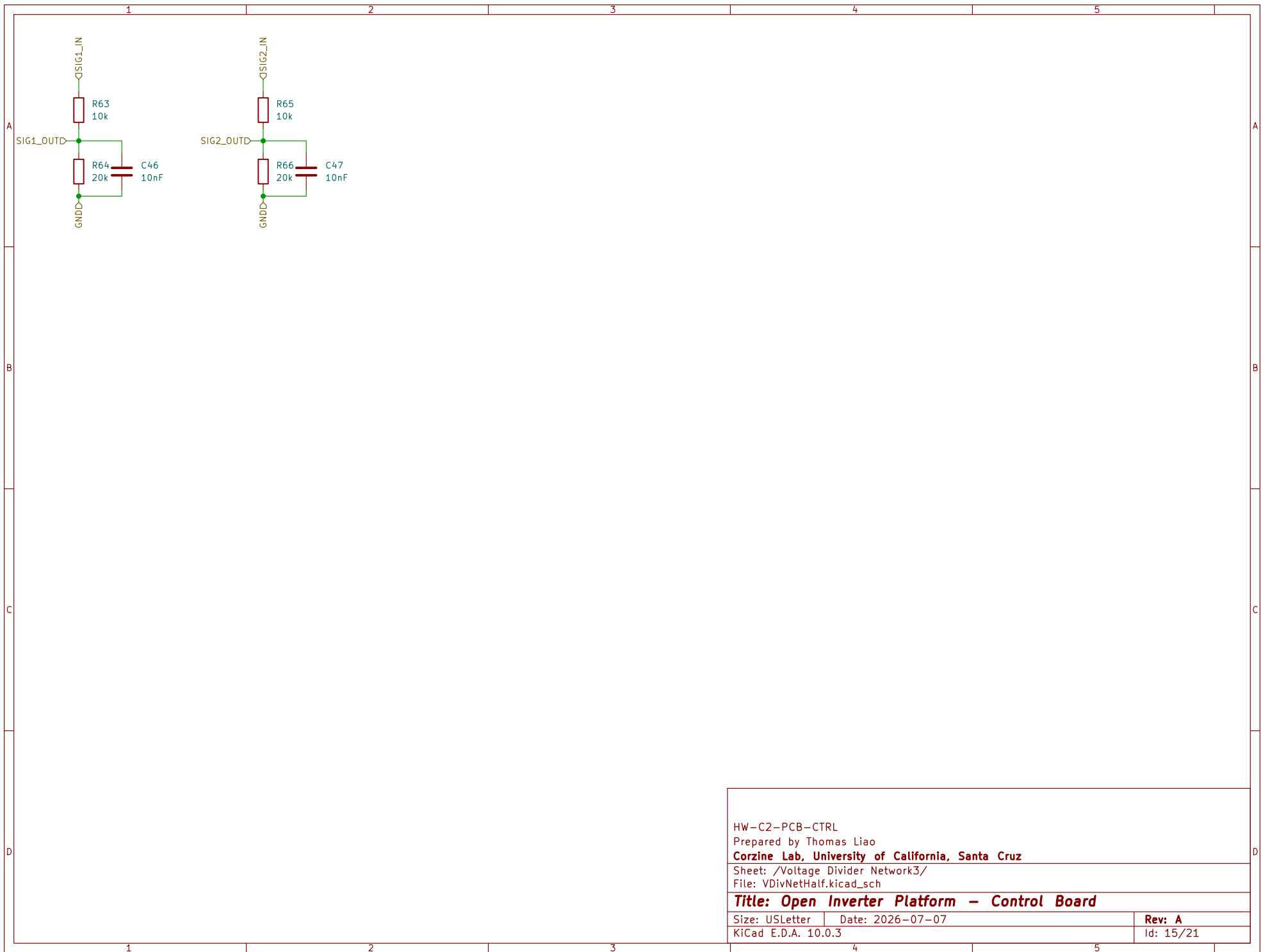
Title: Open Inverter Platform – Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 14/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network3/

File: VDivNetHalf.kicad_sch

Title: Open Inverter Platform – Control Board

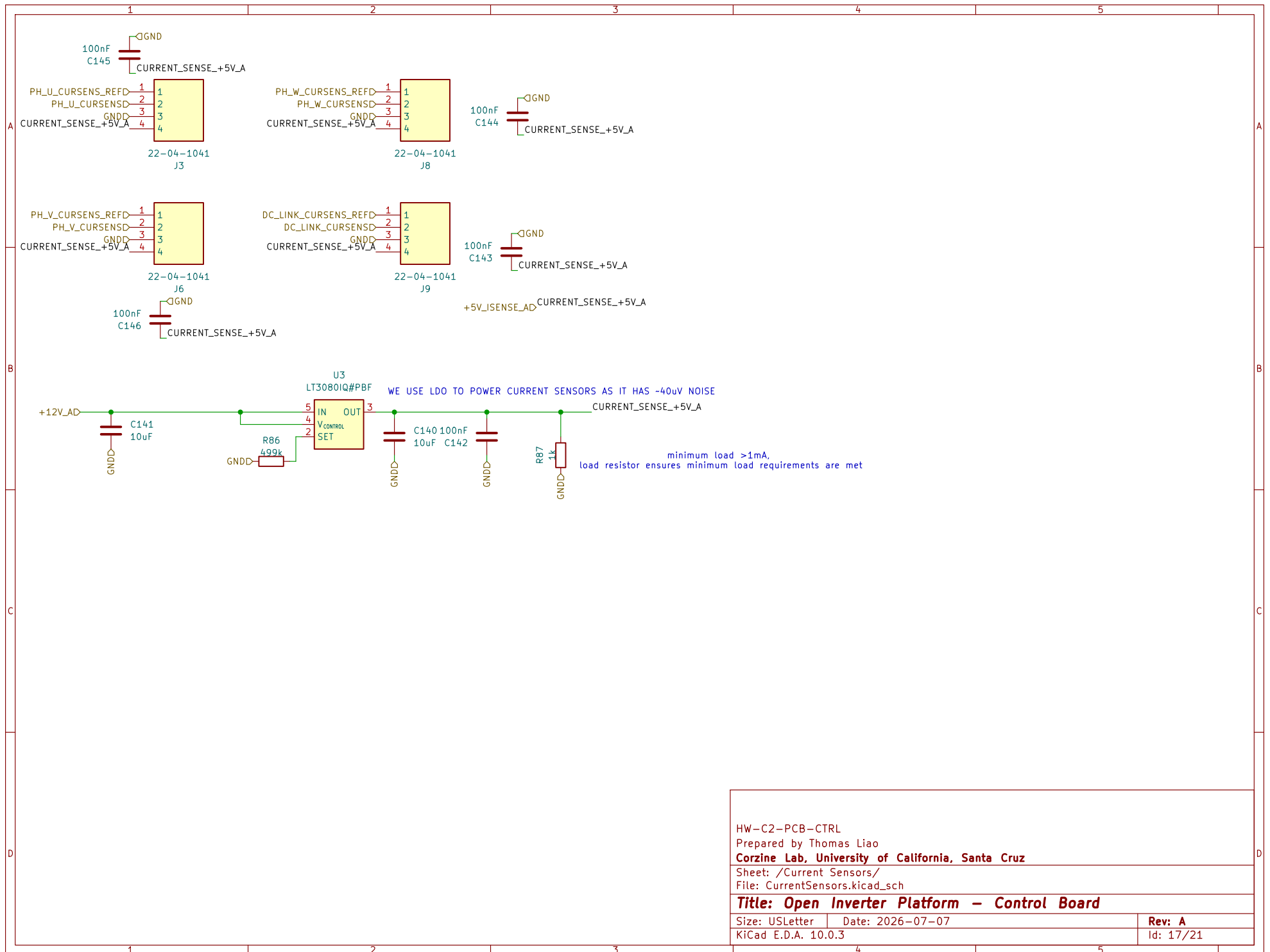
Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 15/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Current Sensors/

File: CurrentSensors.kicad_sch

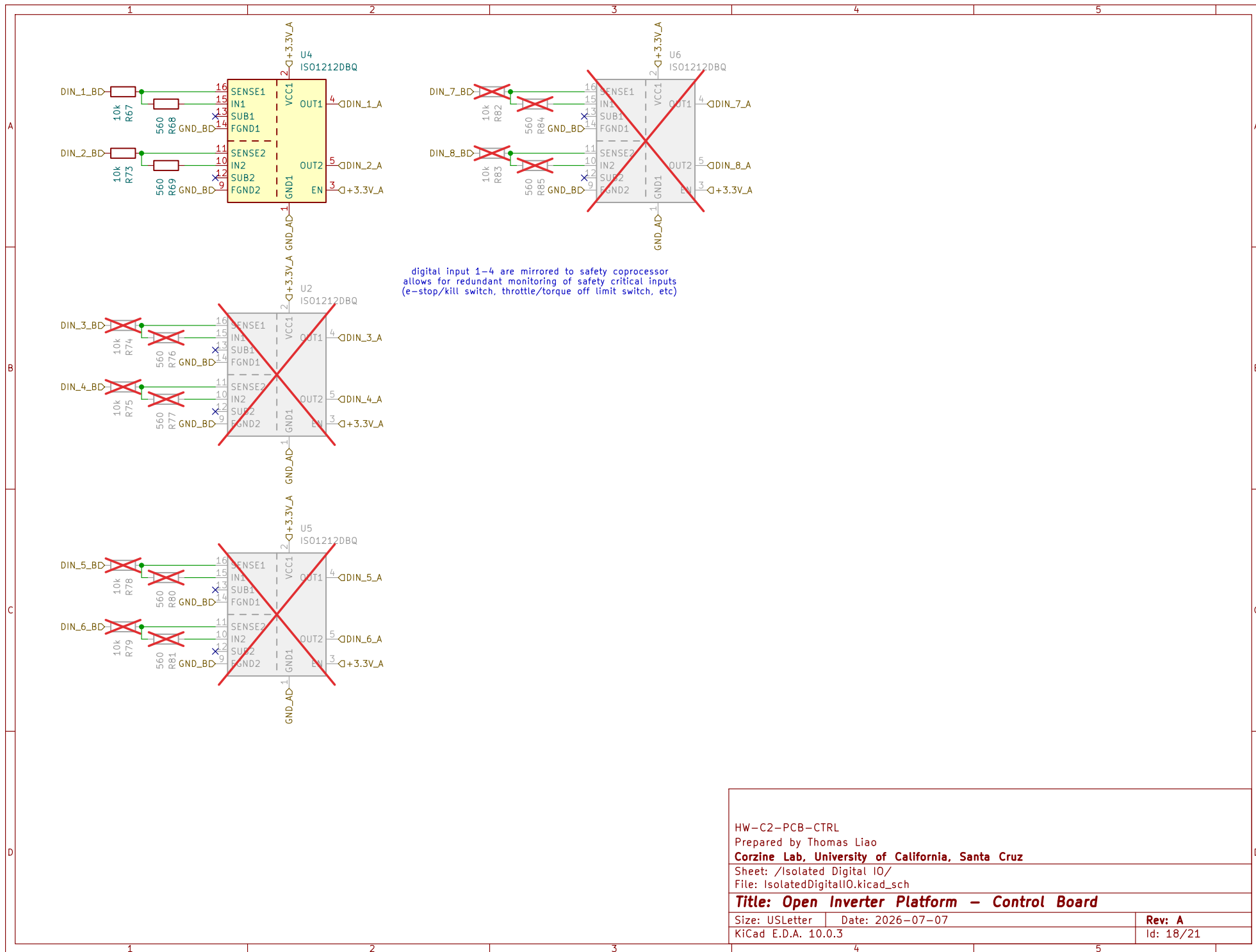
Title: Open Inverter Platform - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 17/21

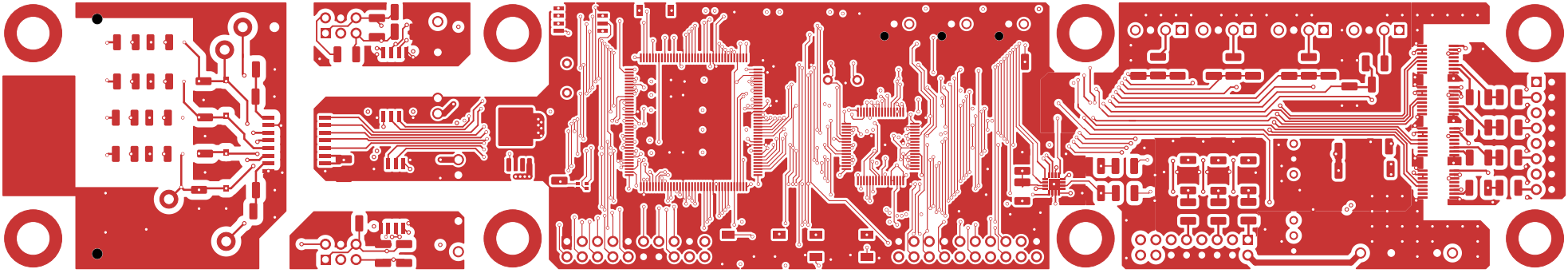


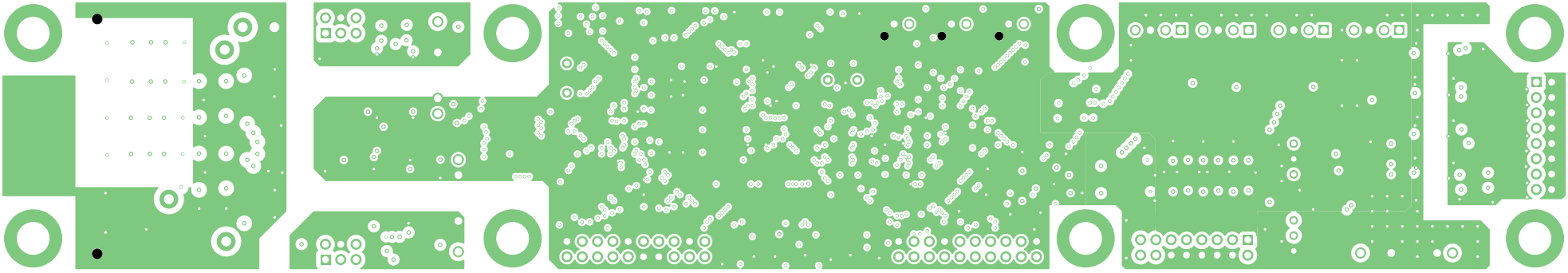
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Isolated Digital IO/
File: IsolatedDigitalIO.kicad_sch

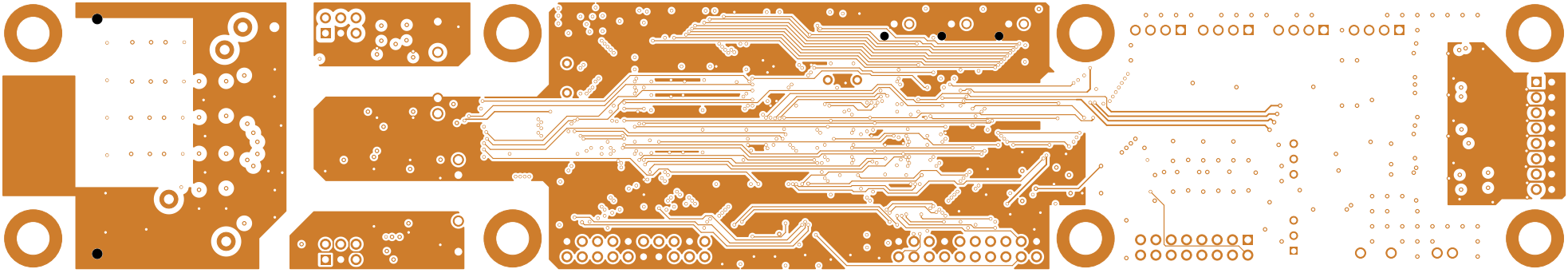
Title: Open Inverter Platform - Control Board

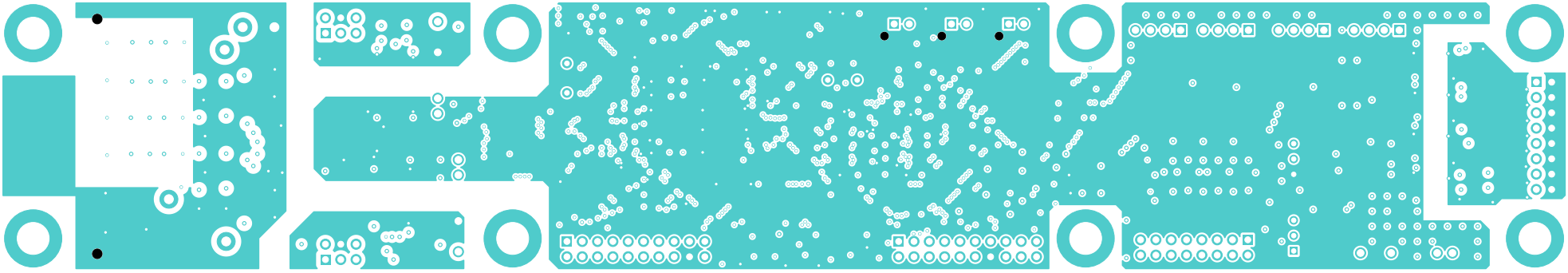
Size: USLetter Date: 2026-07-07
KiCad E.D.A. 10.0.3

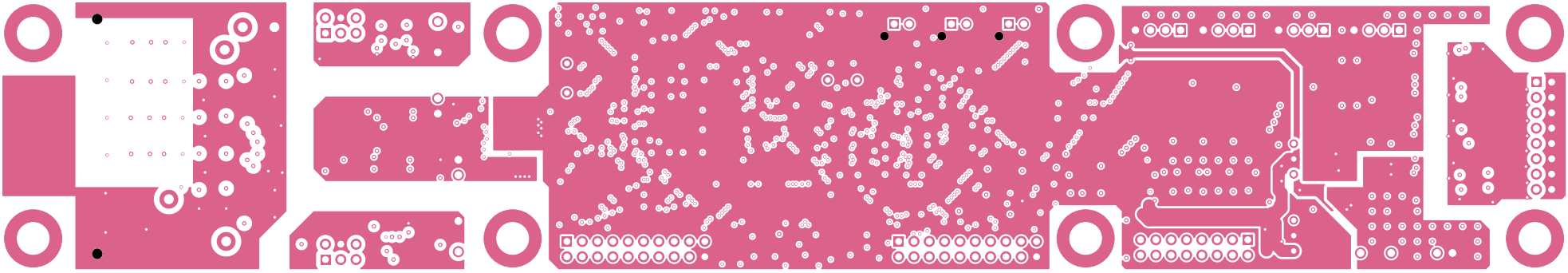
Rev: A
Id: 18/21

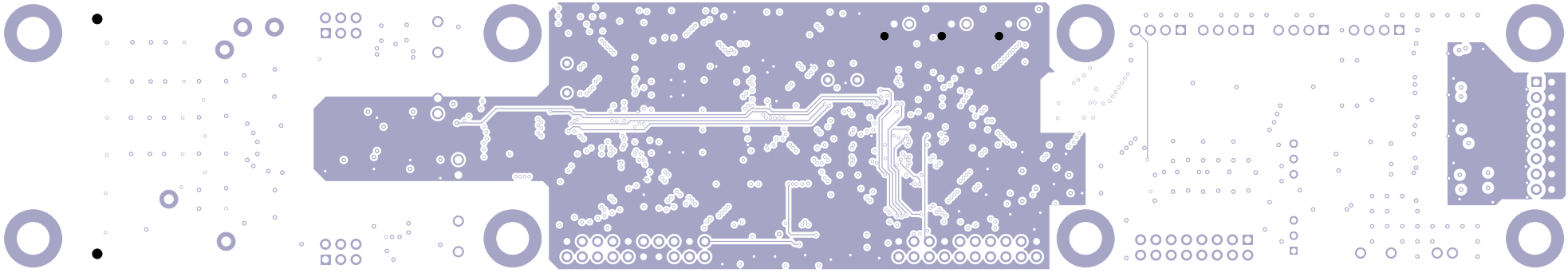


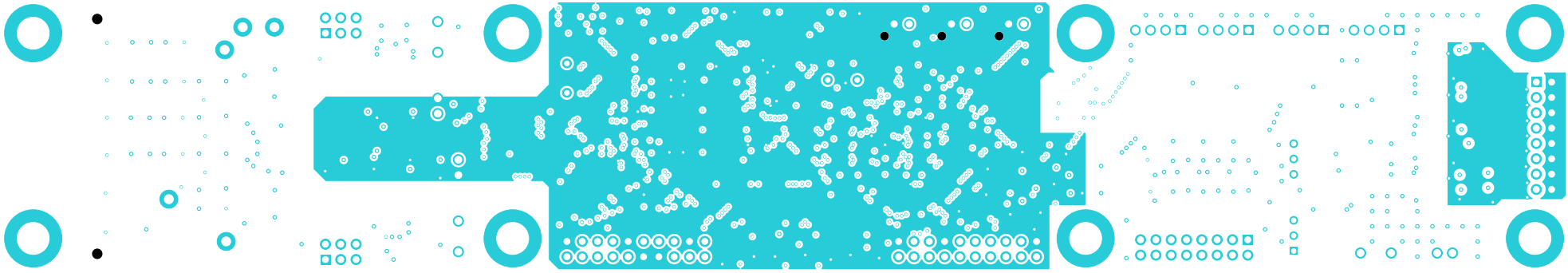


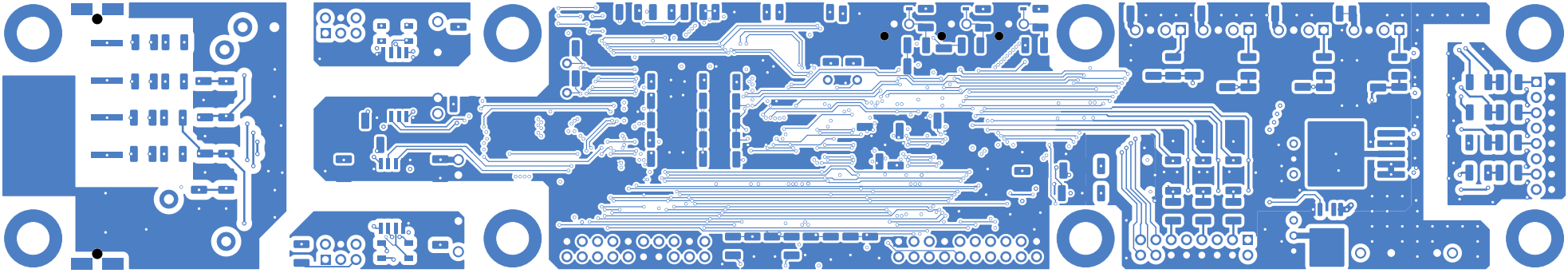


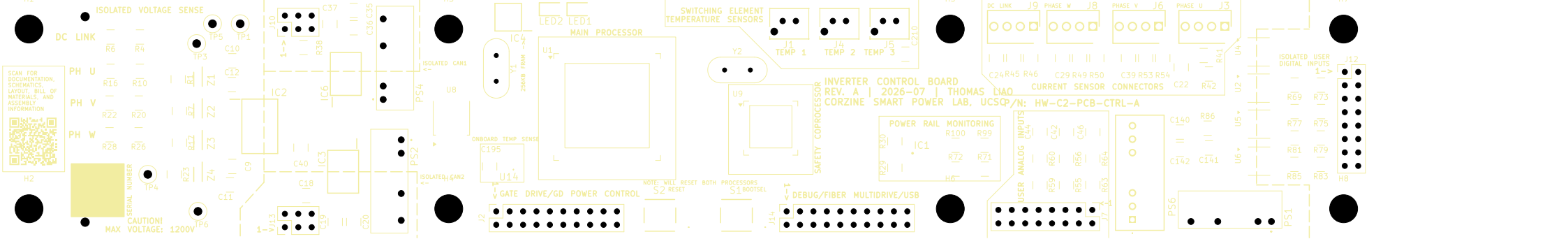






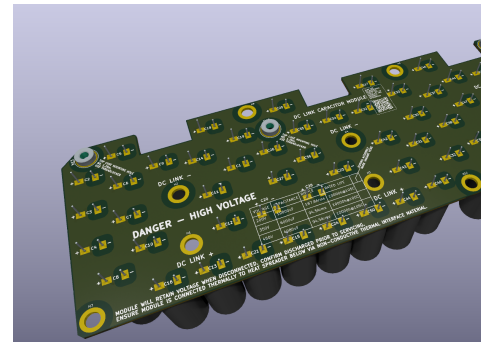
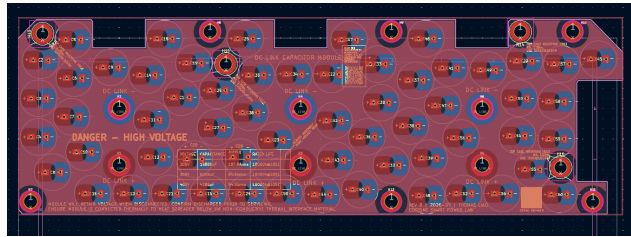
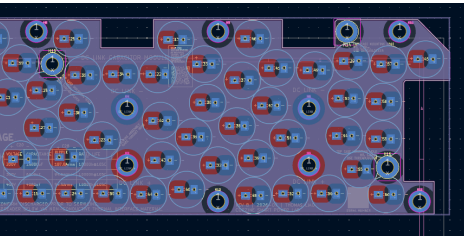


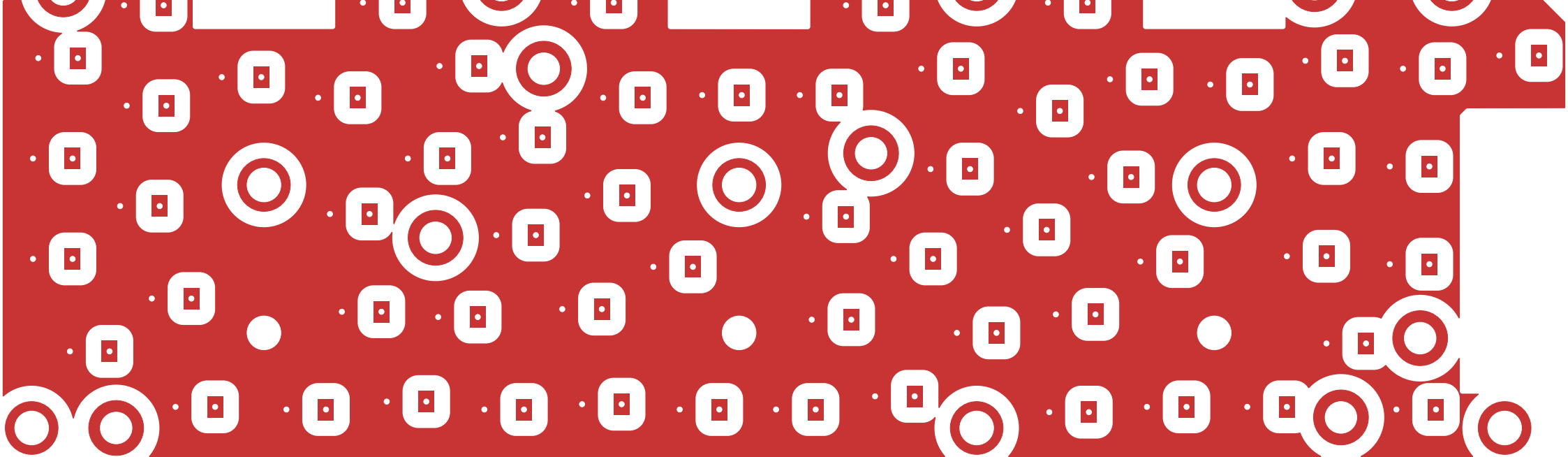


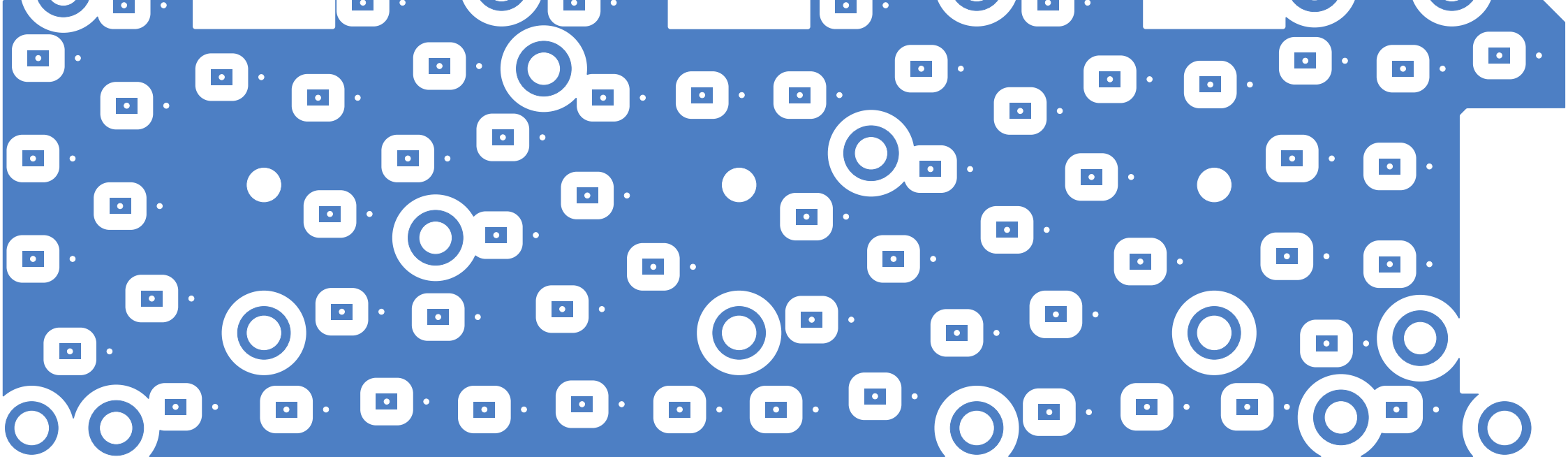


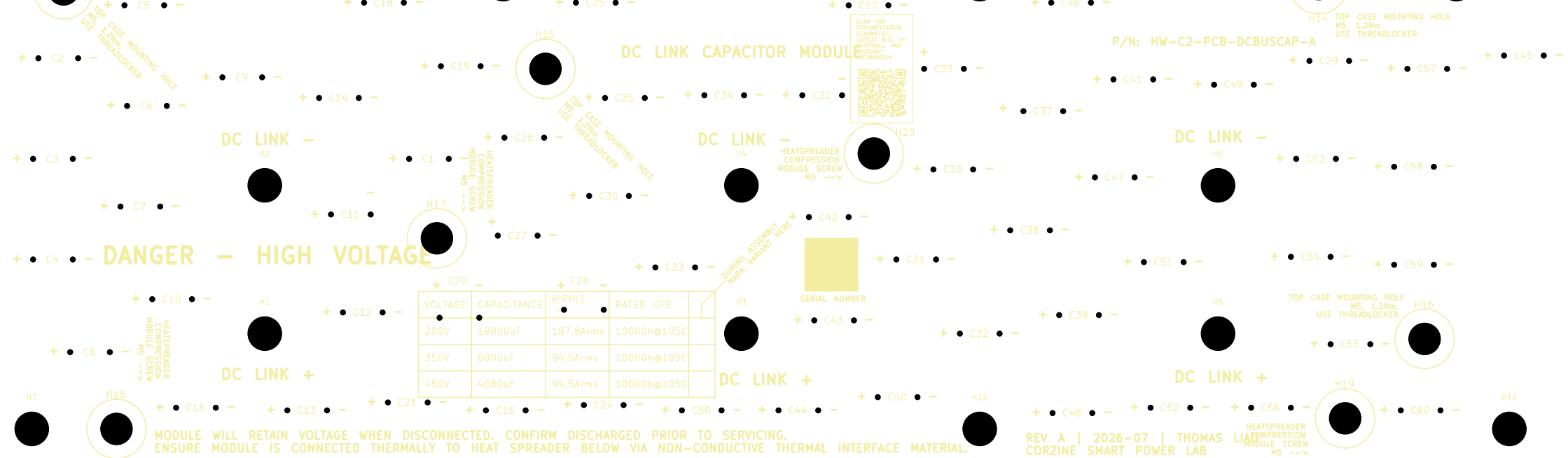
DCBUSCAPACITORBOARD

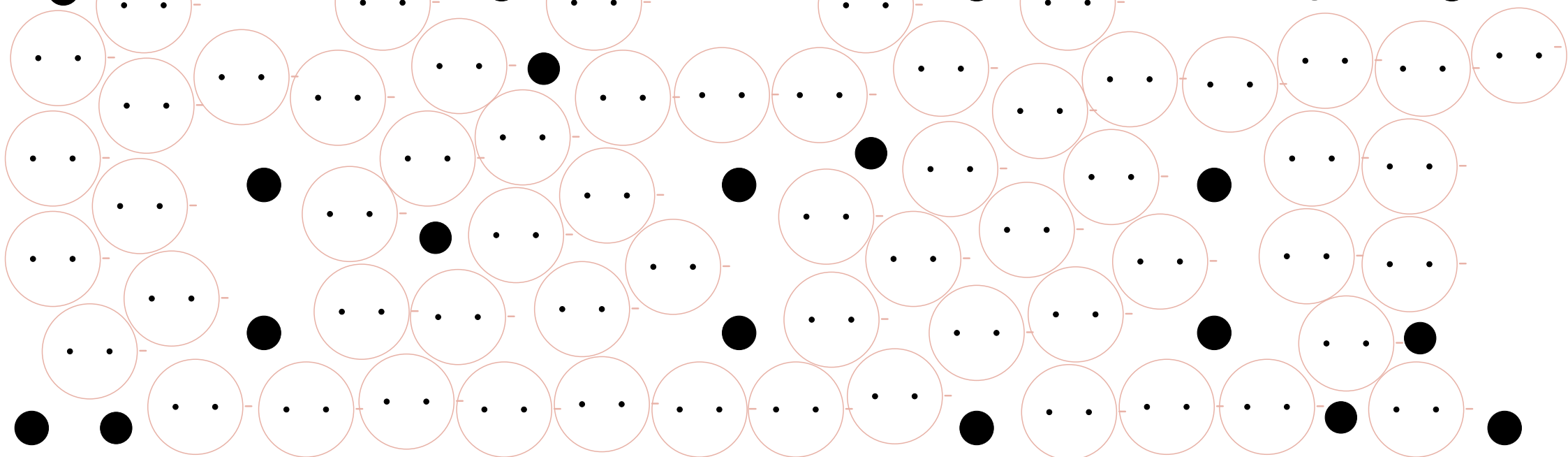
HW-C2-PCB-DCBUSCAP-A · printed circuit board — schematic and PCB layers





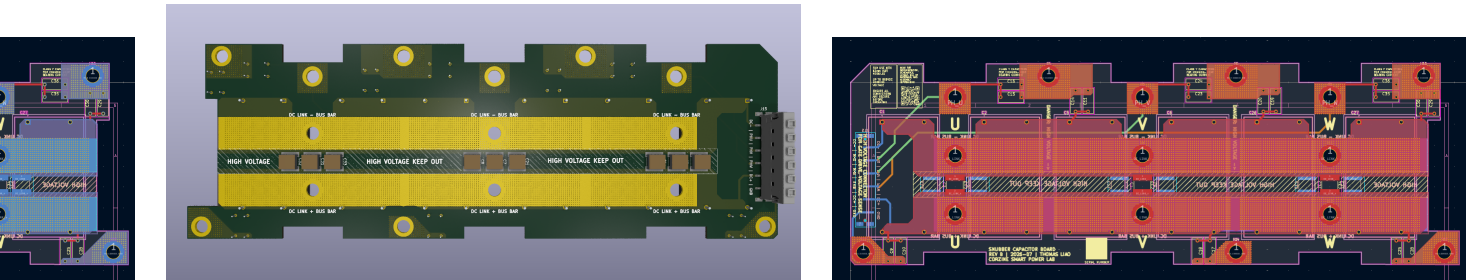


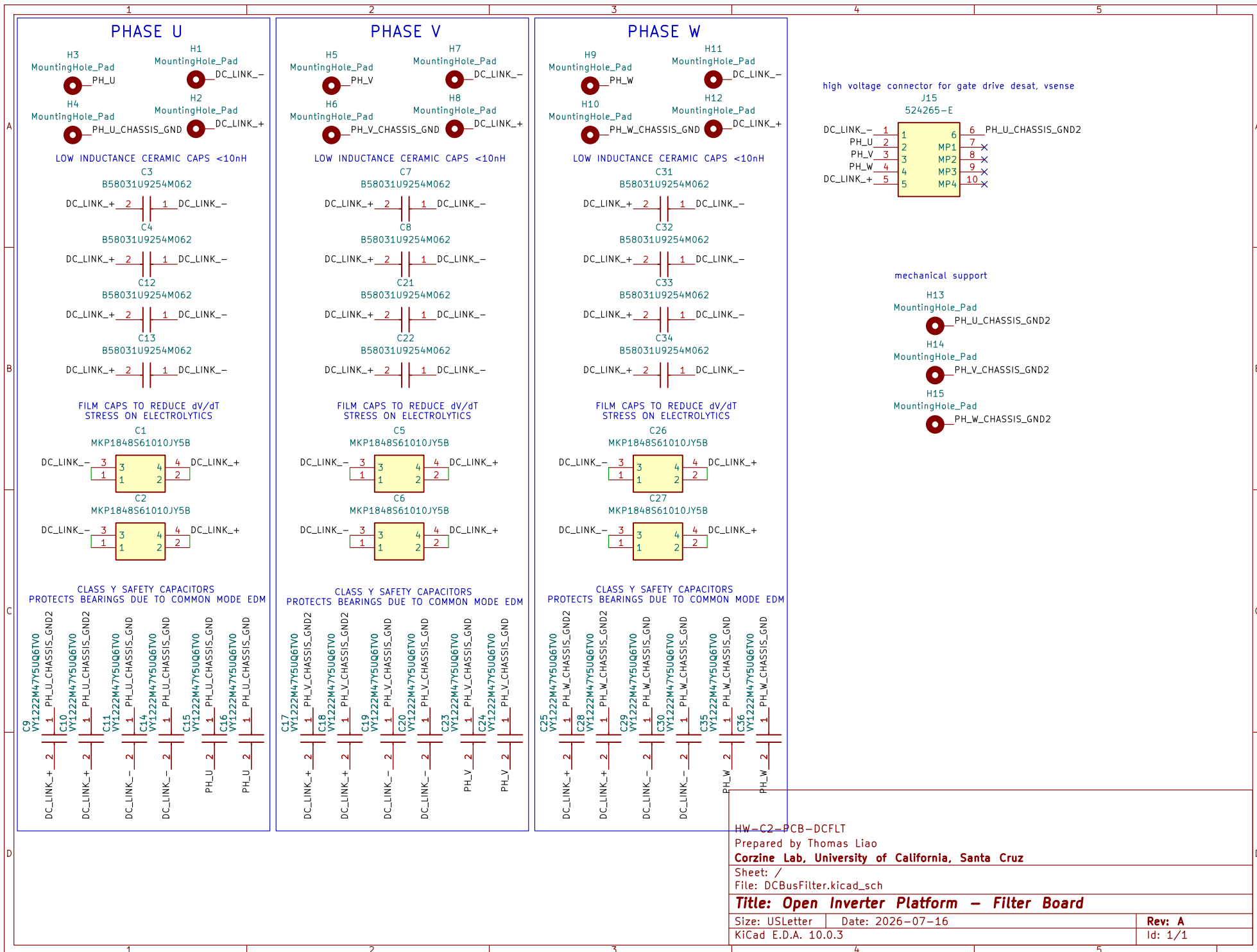


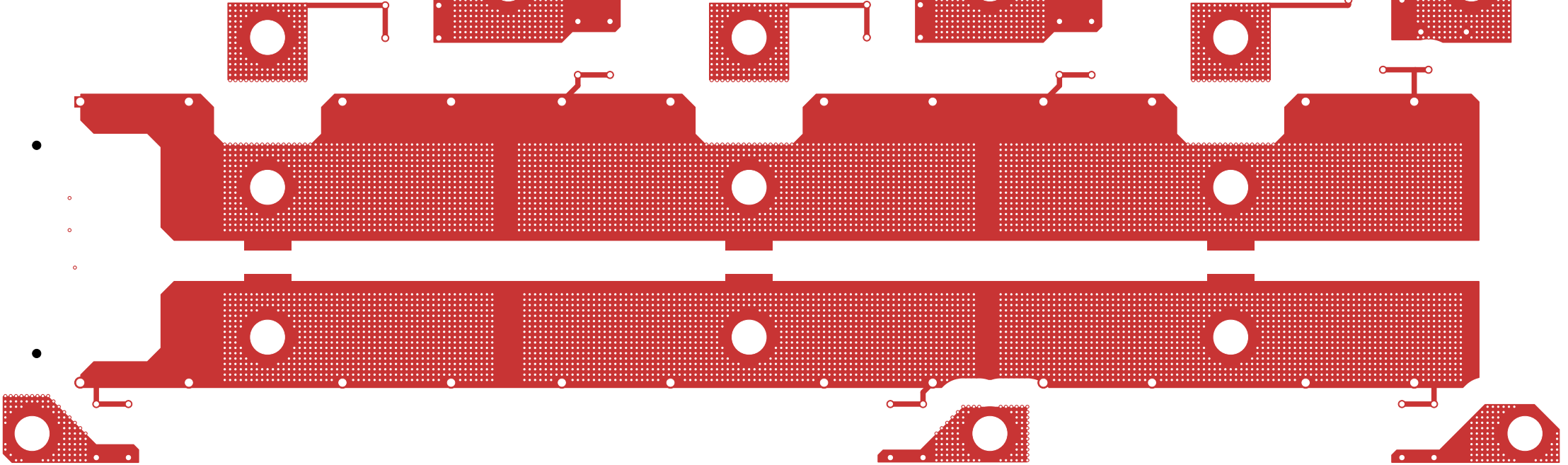


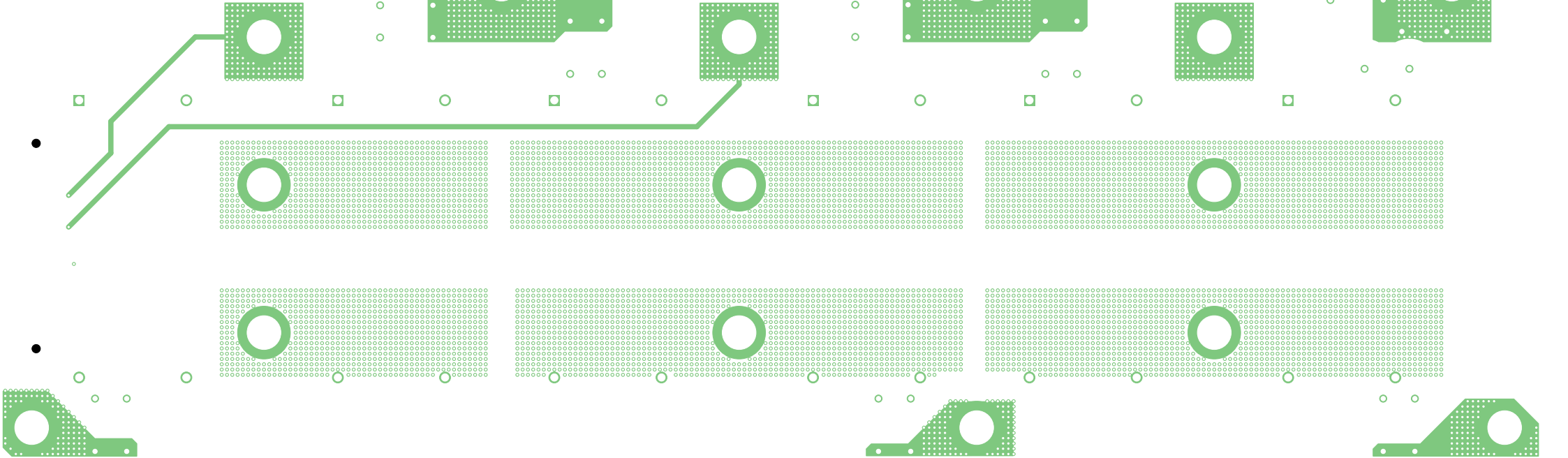
DCBUSFILTER

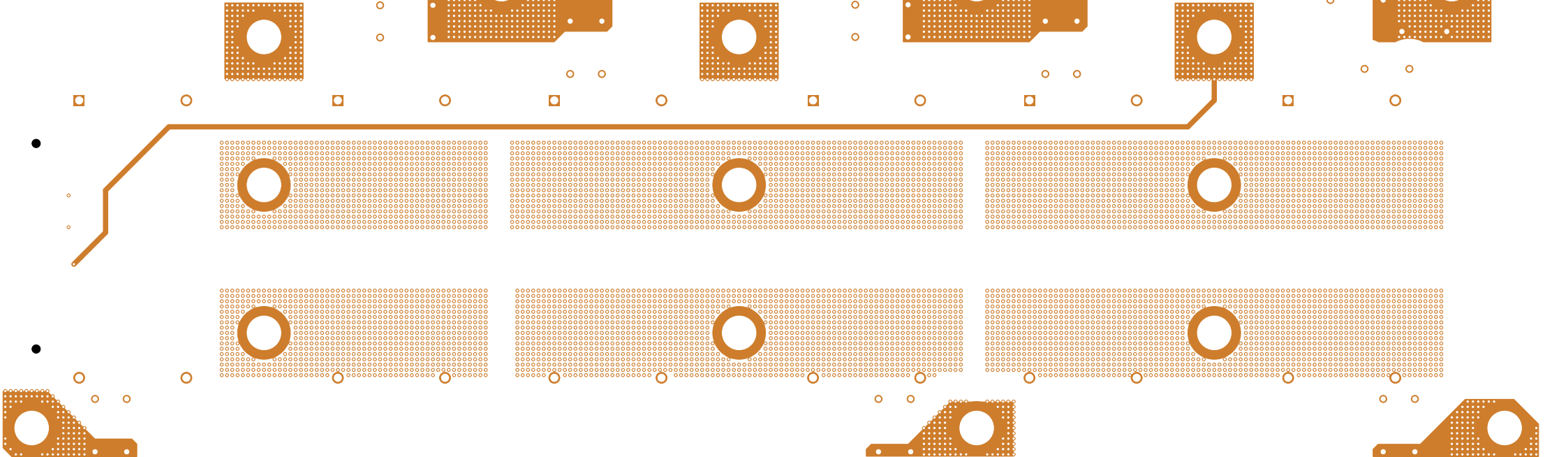
HW-C2-PCB-DCFLT-A · printed circuit board — schematic and PCB layers

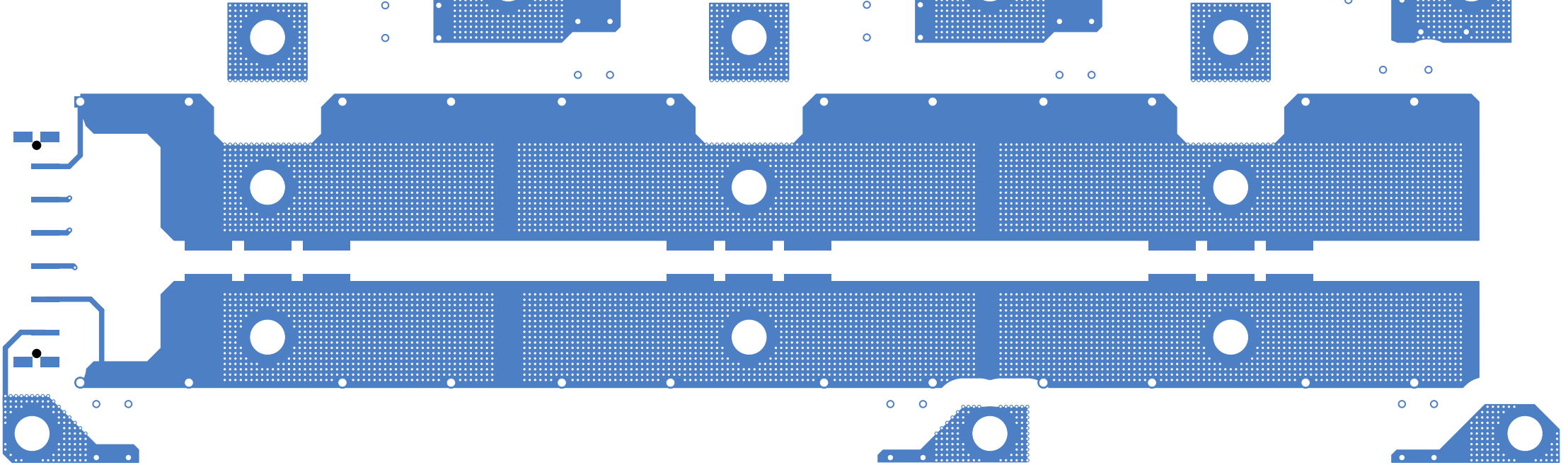


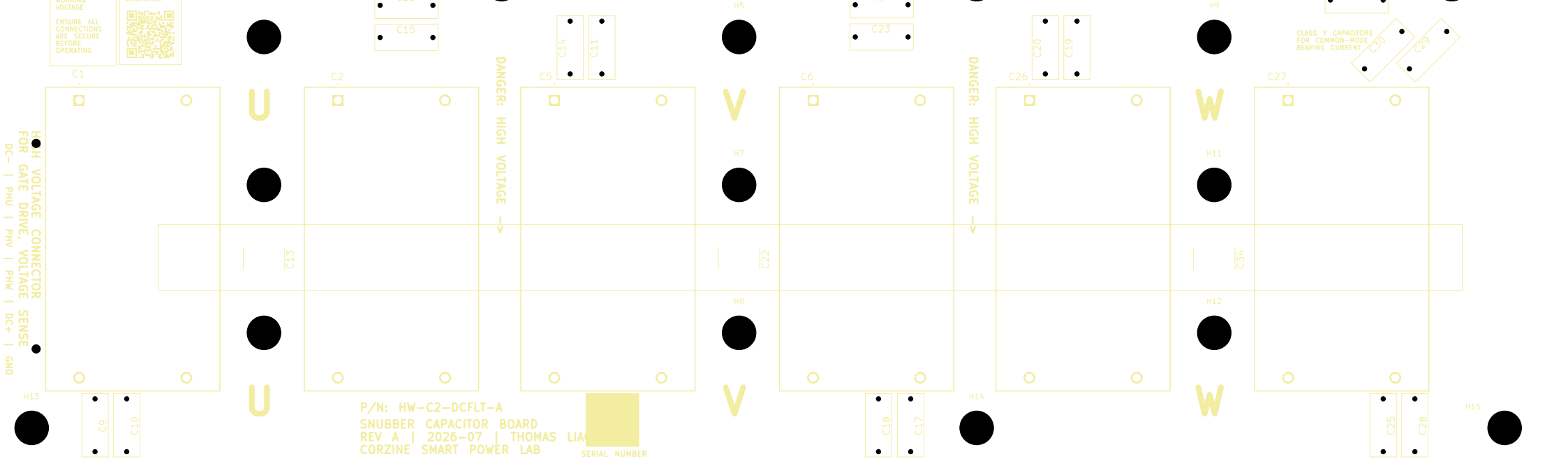


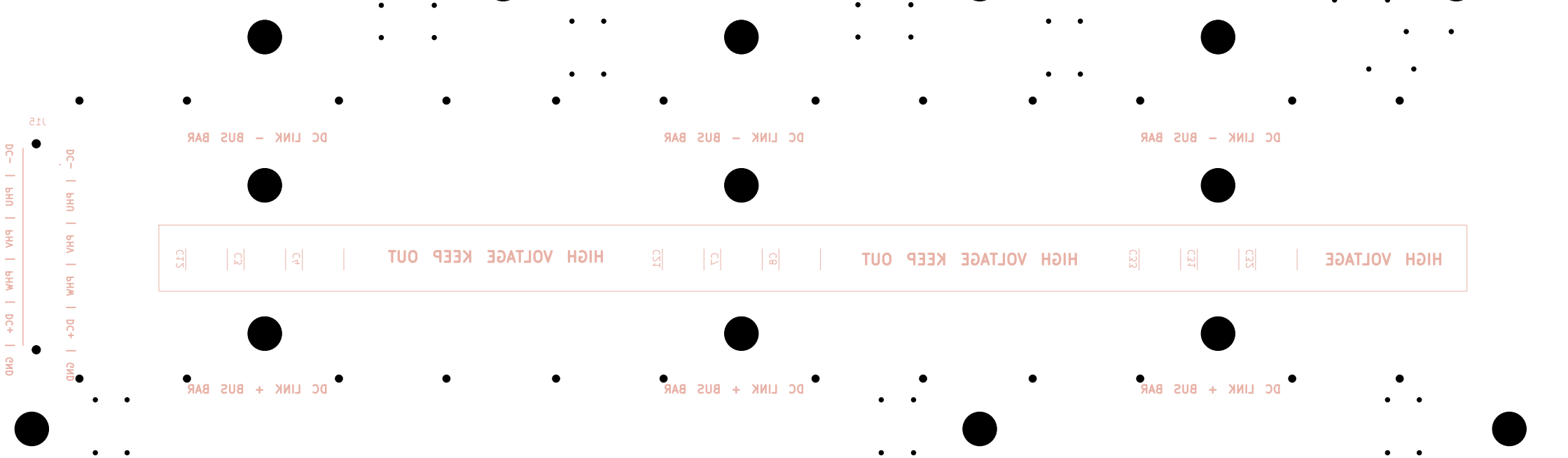






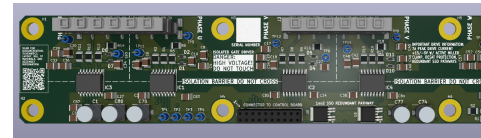
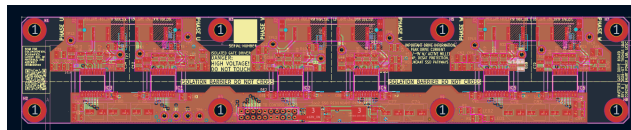
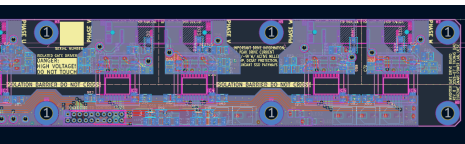


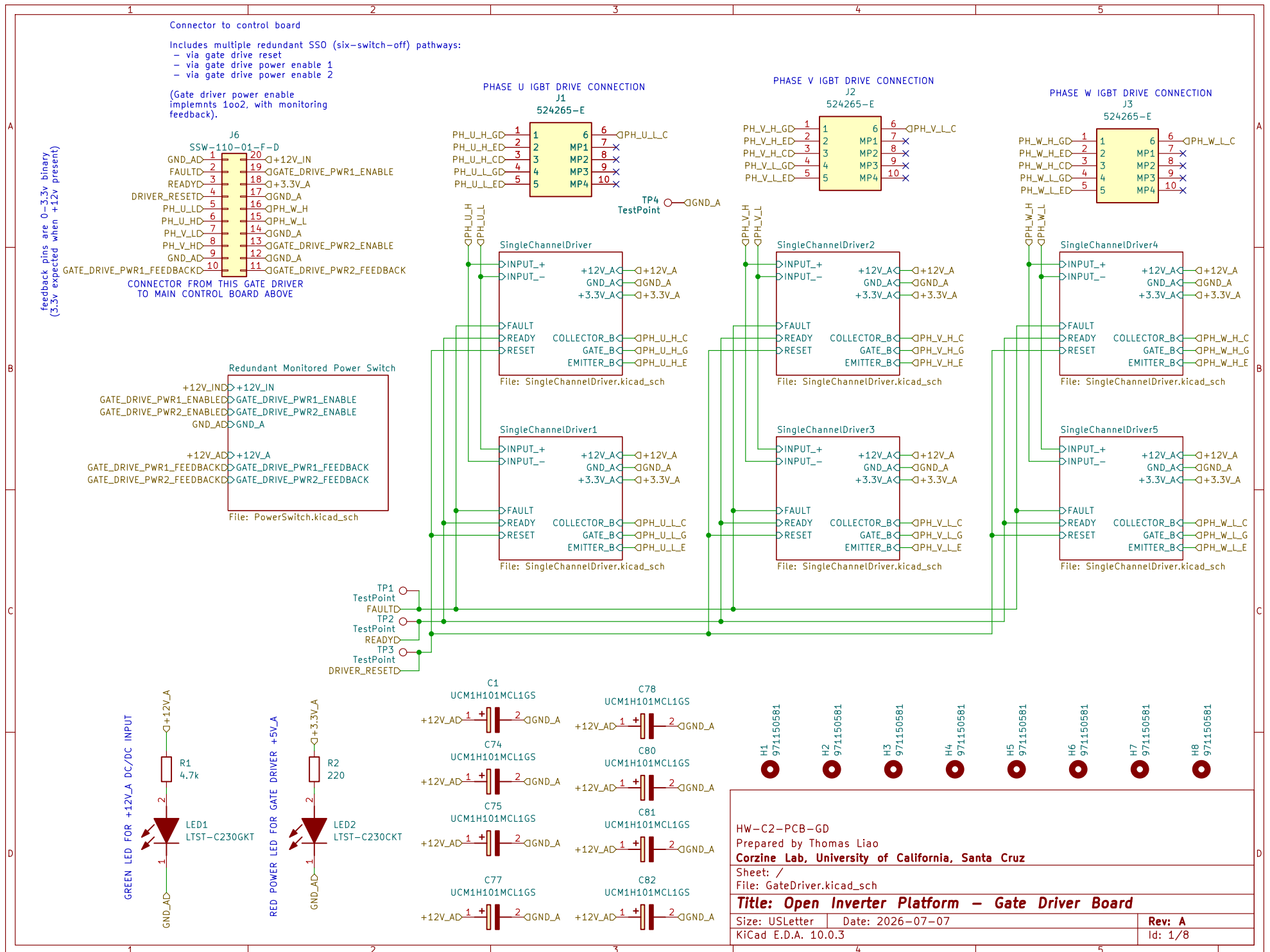


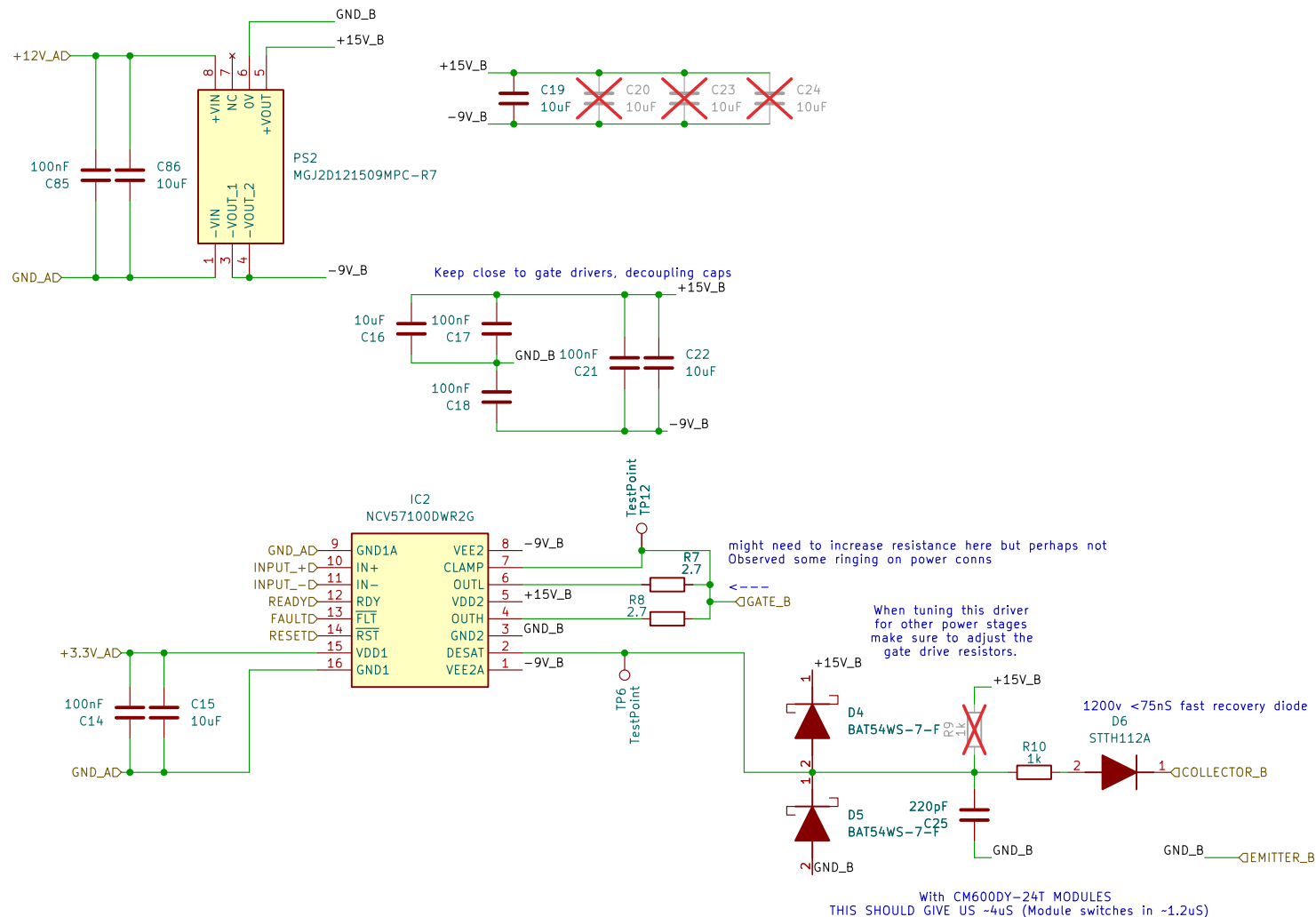


GATEDRIVER

HW-C2-PCB-GD-A · printed circuit board — schematic and PCB layers







HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver2/

File: SingleChannelDriver.kicad_sch

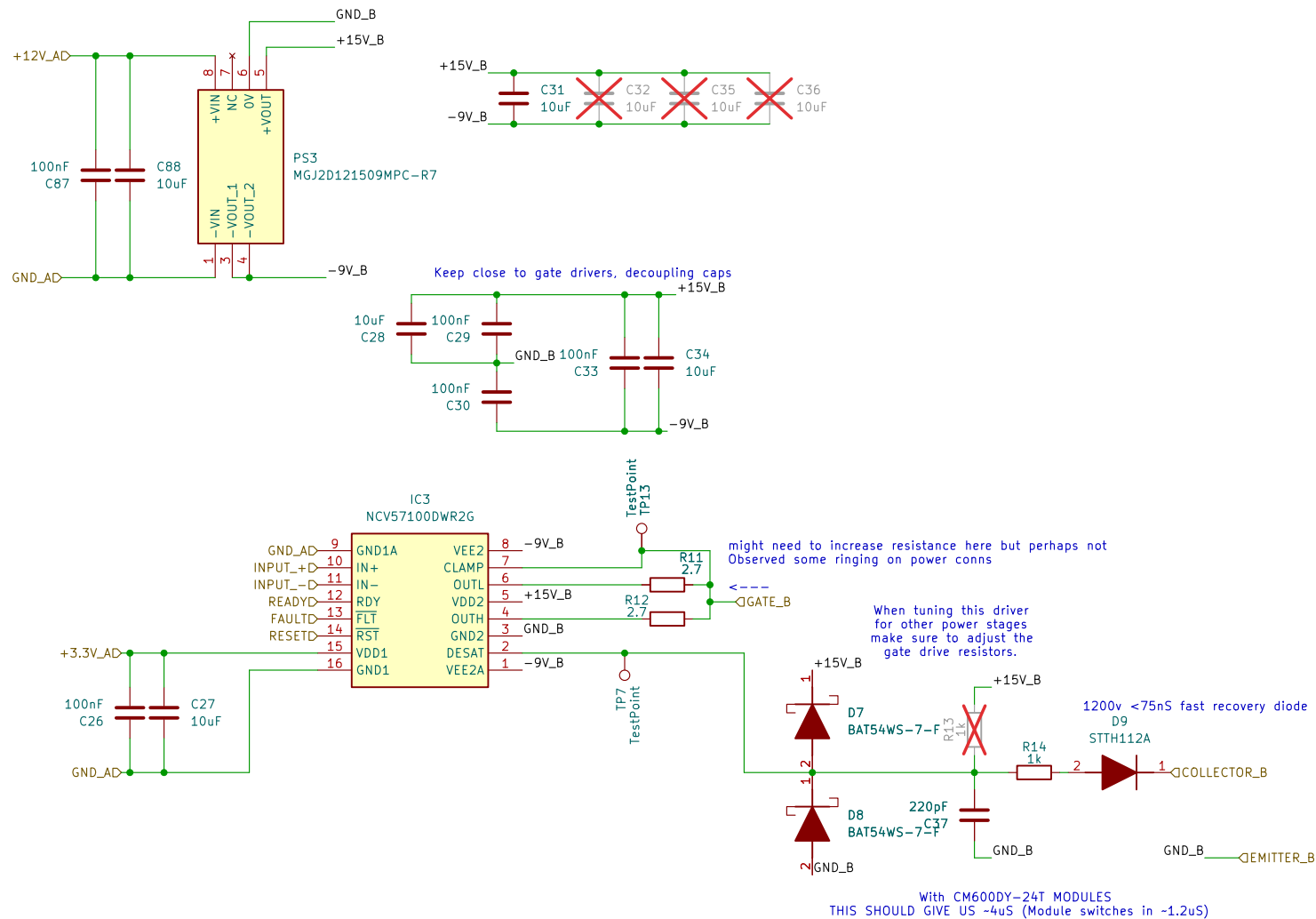
Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 3/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver/

File: SingleChannelDriver.kicad_sch

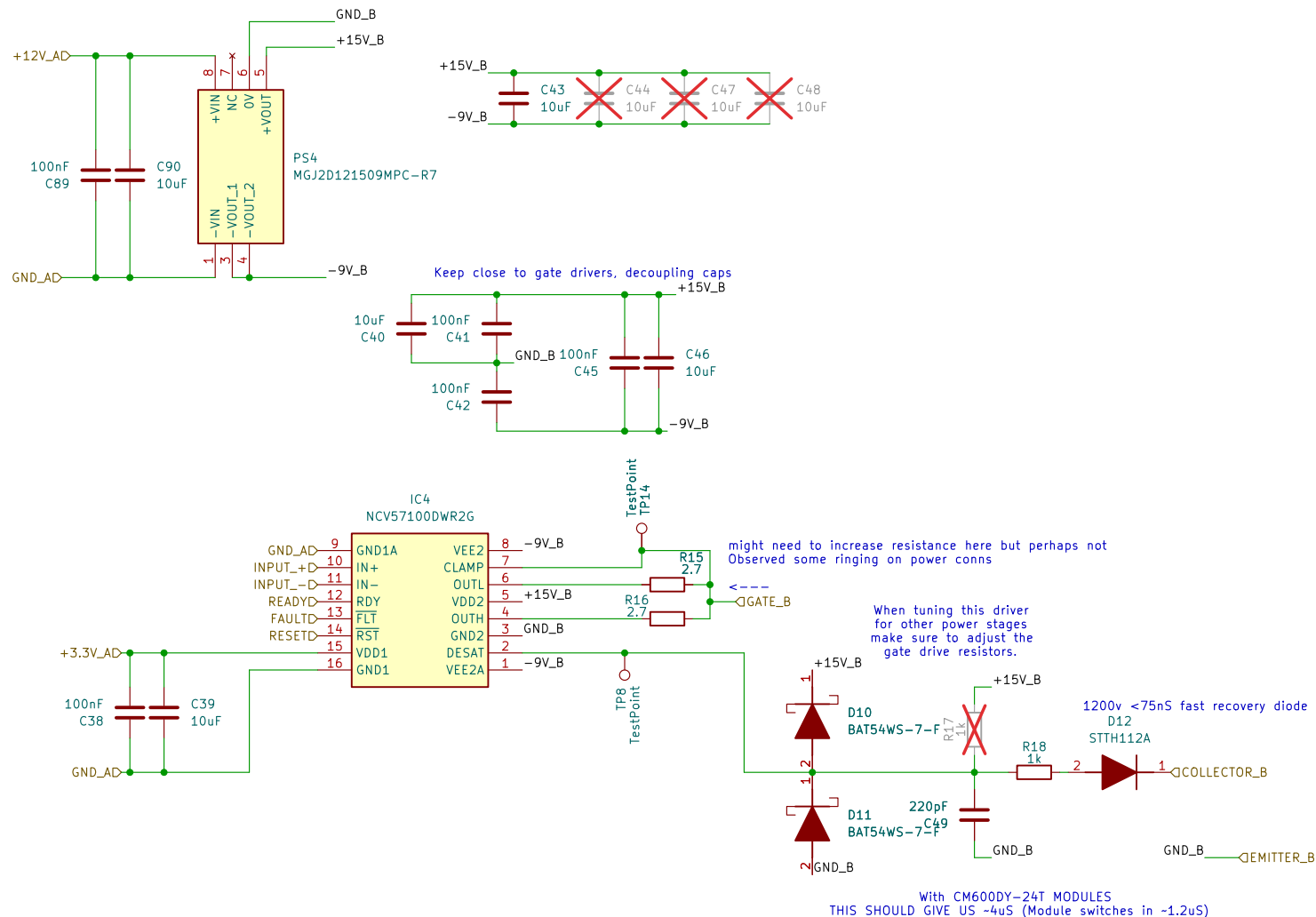
Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 4/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver3/

File: SingleChannelDriver.kicad_sch

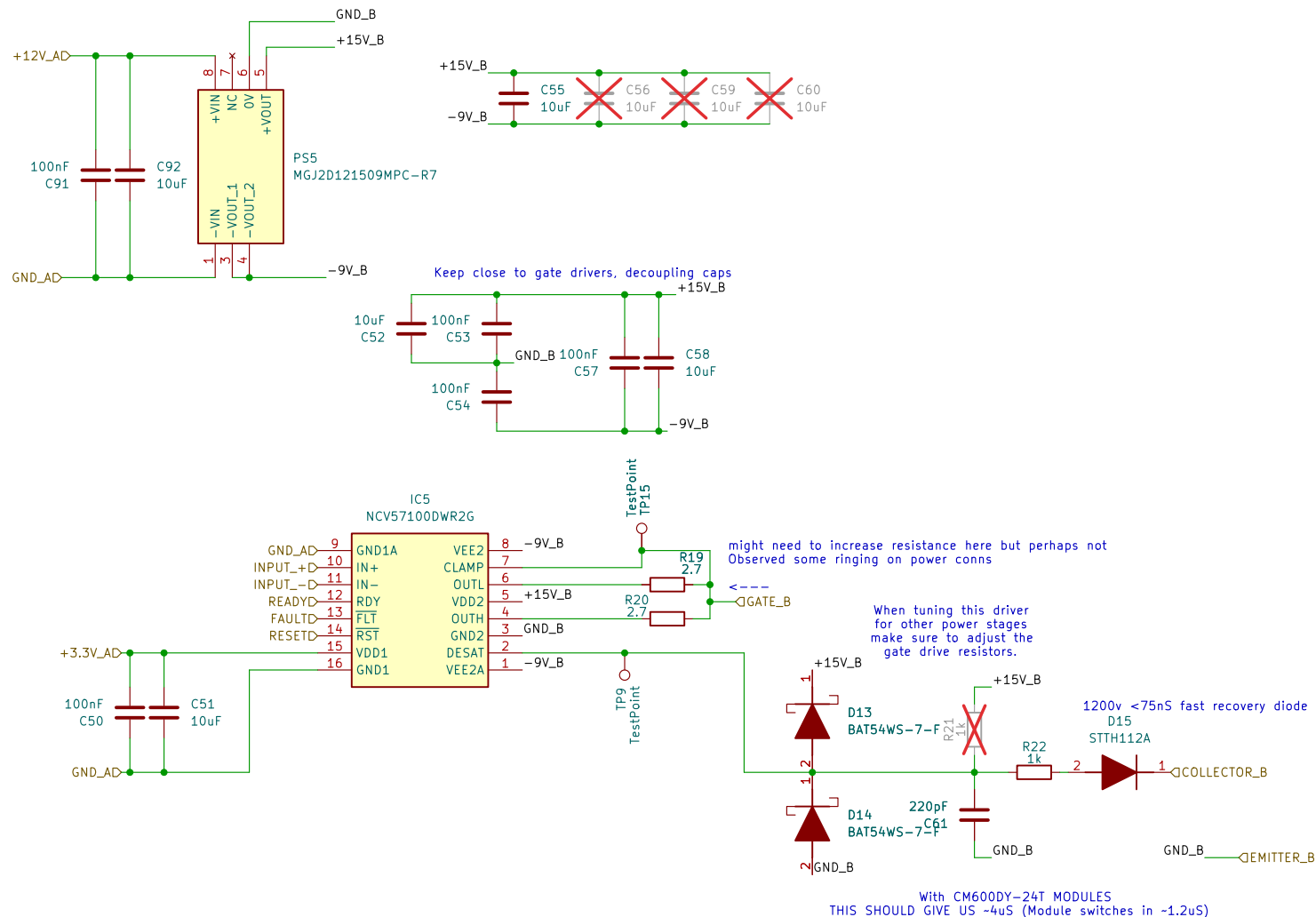
Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 5/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver4/

File: SingleChannelDriver.kicad_sch

Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

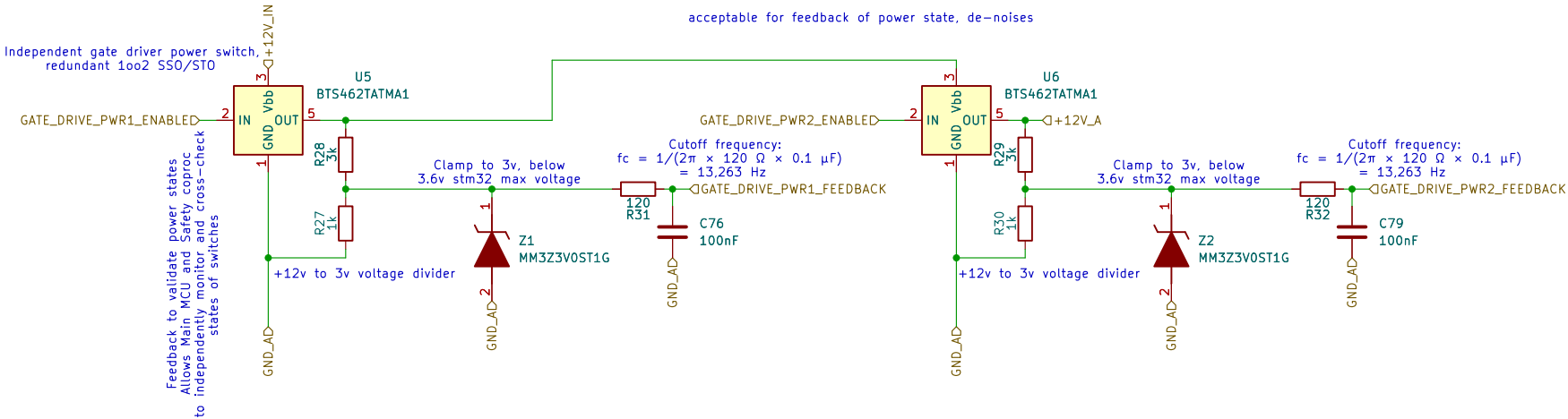
Id: 6/8

This circuit implements a redundant pathway for both mcus to implement Six Switch Open / Safe Torque Off in addition to regular pwm break and gate driver reset.

The following switches are independent systems with feedback monitoring. Each MCU cross monitors returned inputs, and can validate if the switch failed to respond as commanded – thus removing latent failures from this safety subsystem.

cutoff period (1/freq) ->
75.4 uS

acceptable for feedback of power state, de-noises



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Redundant Monitored Power Switch/

File: PowerSwitch.kicad_sch

Title: Open Inverter Platform – Gate Driver Board

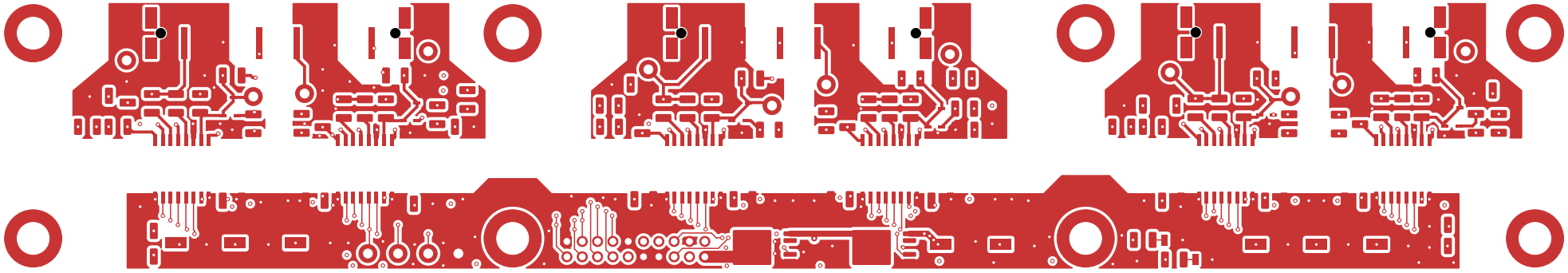
Size: USLetter

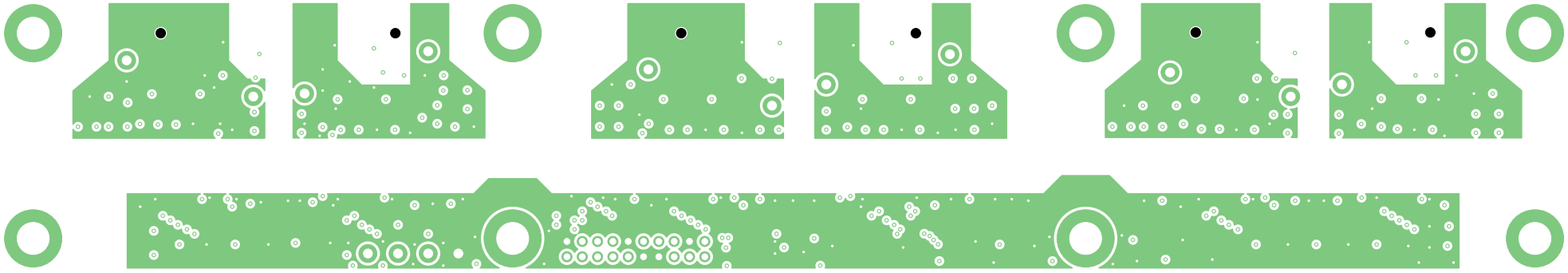
Date: 2026-07-07

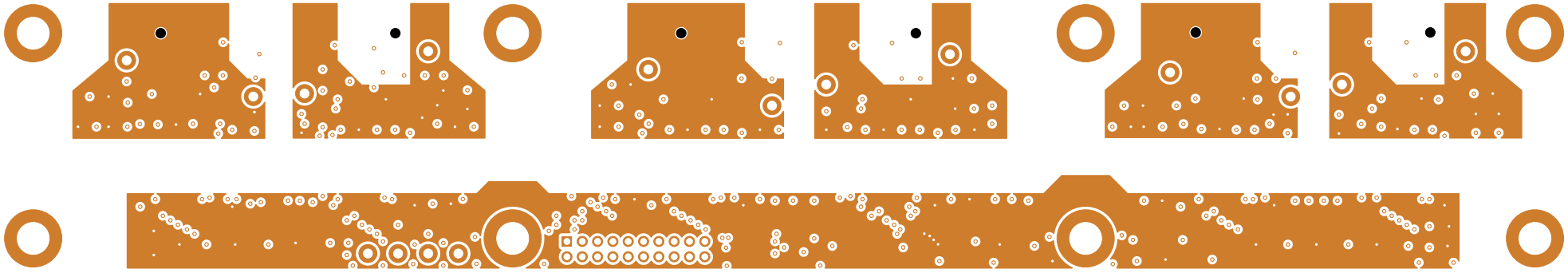
Rev: A

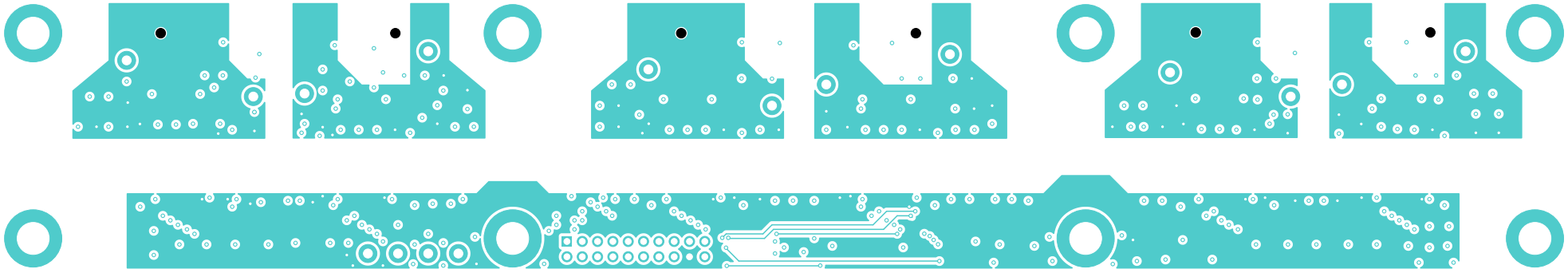
KiCad E.D.A. 10.0.3

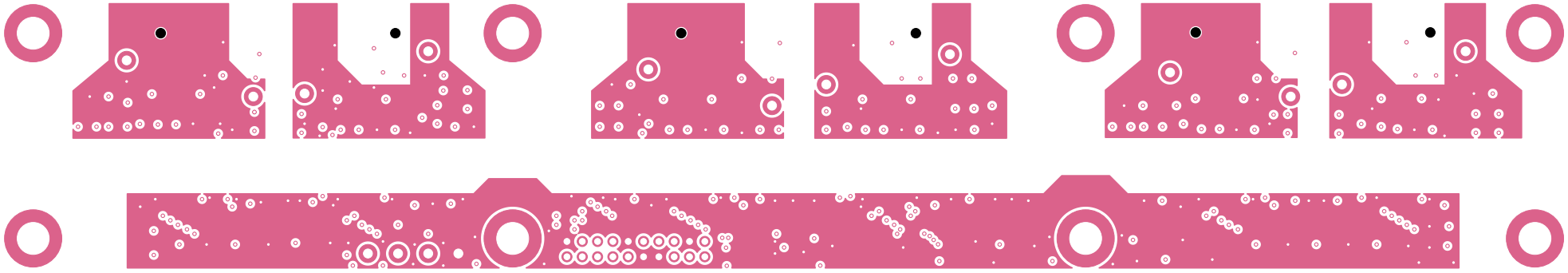
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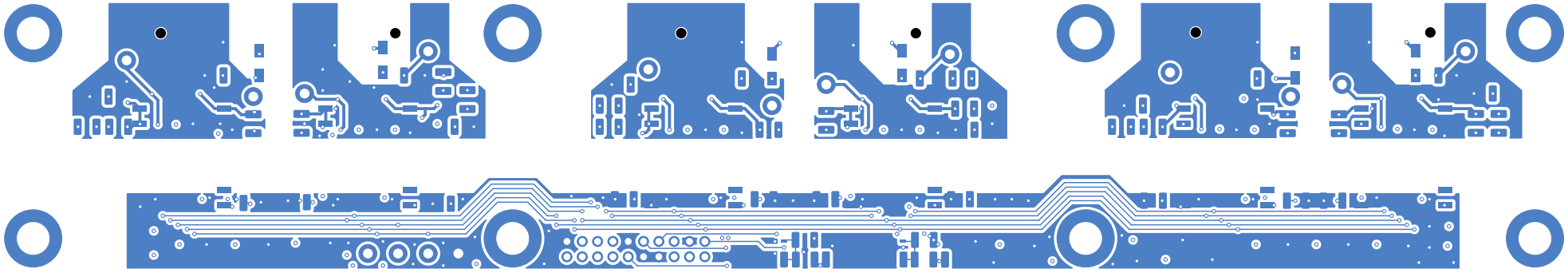


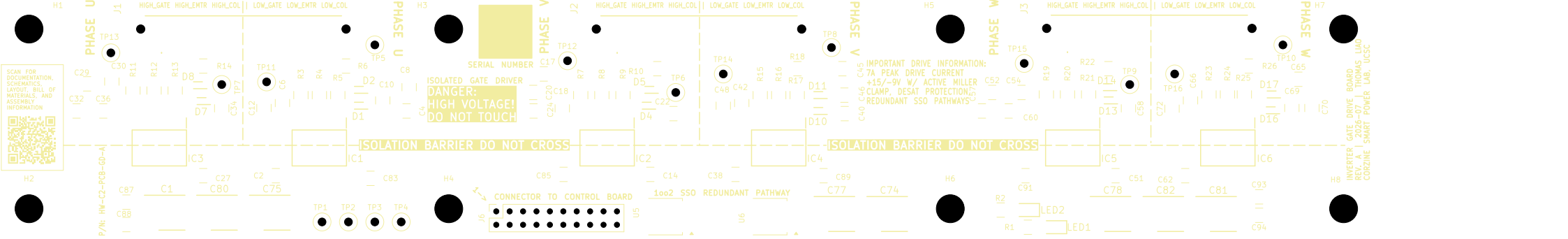


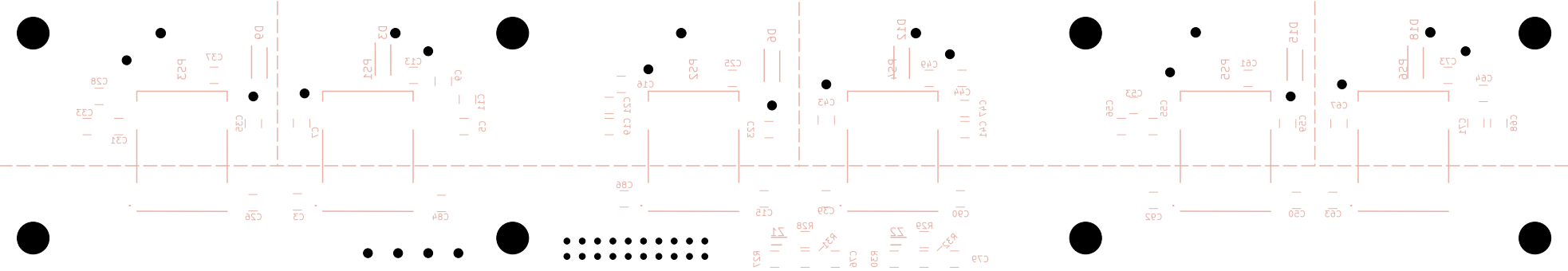






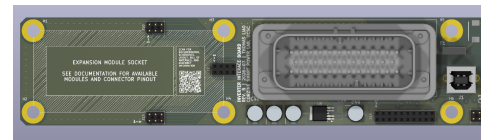
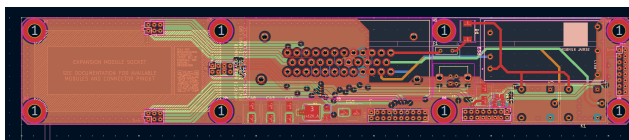
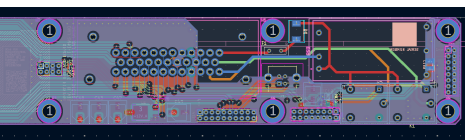


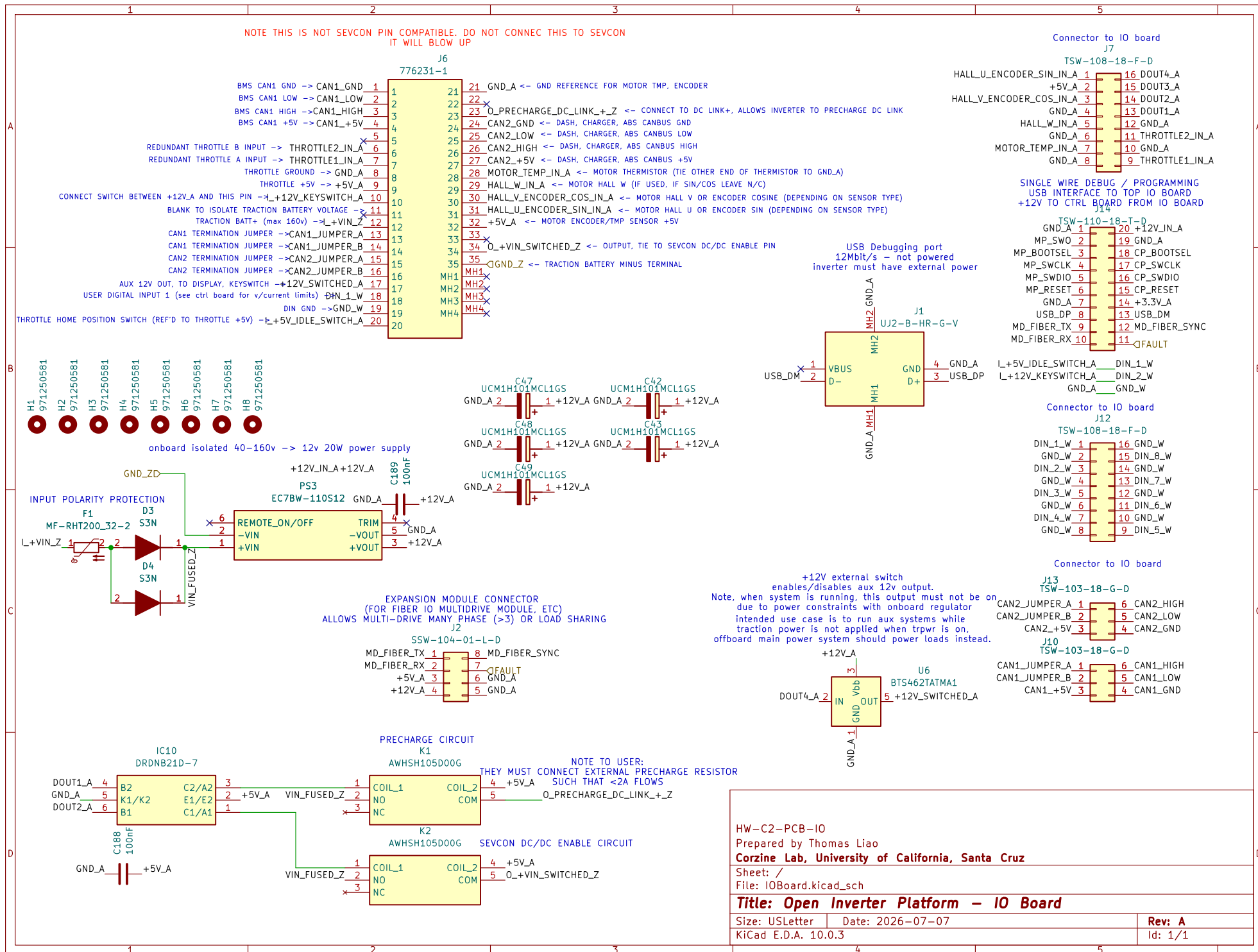


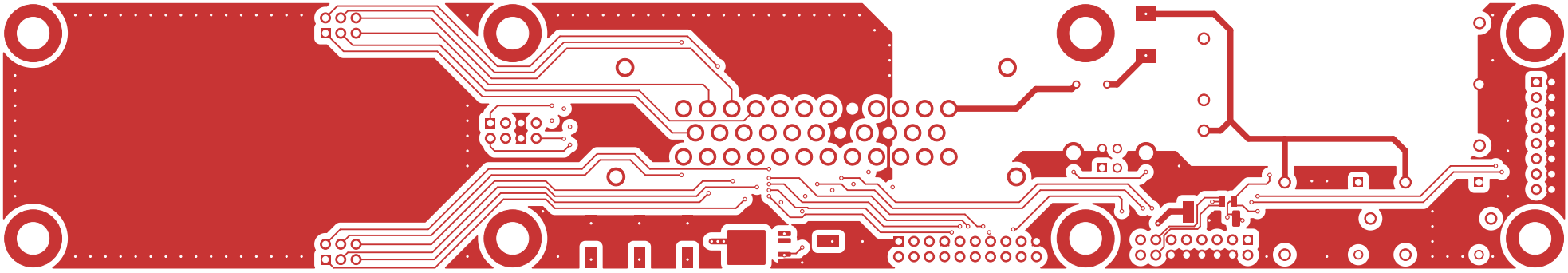


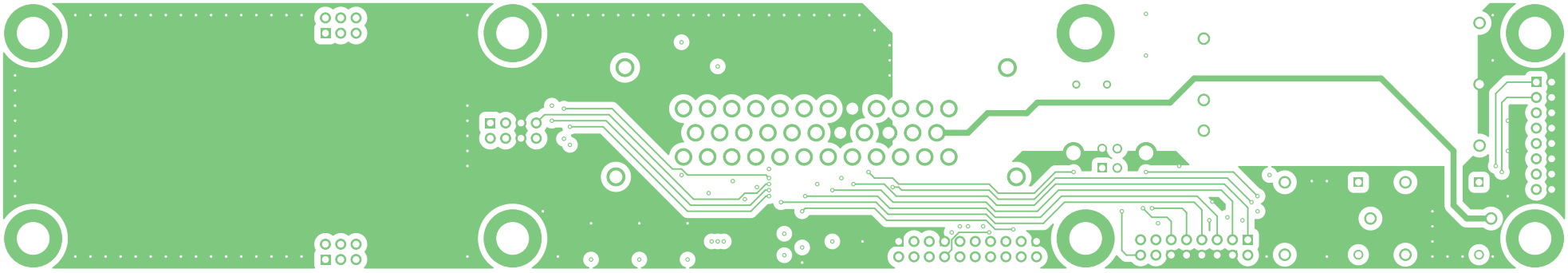
IOBOARD

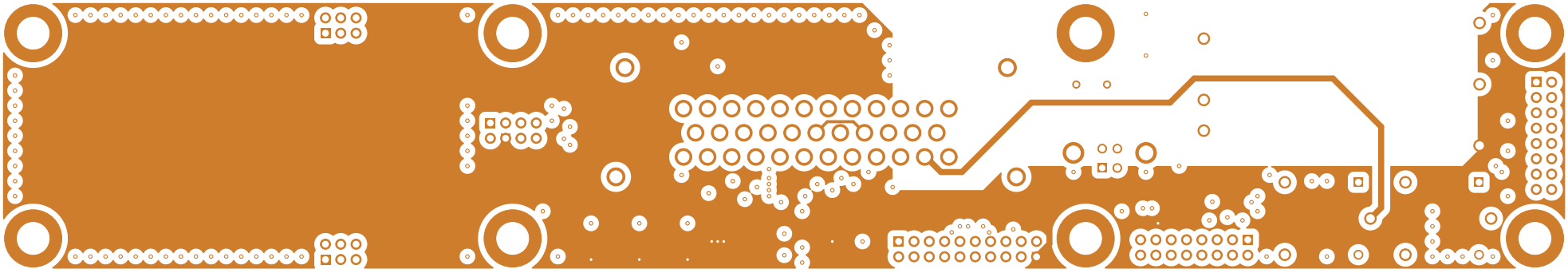
HW-C2-PCB-IO-A · printed circuit board — schematic and PCB layers

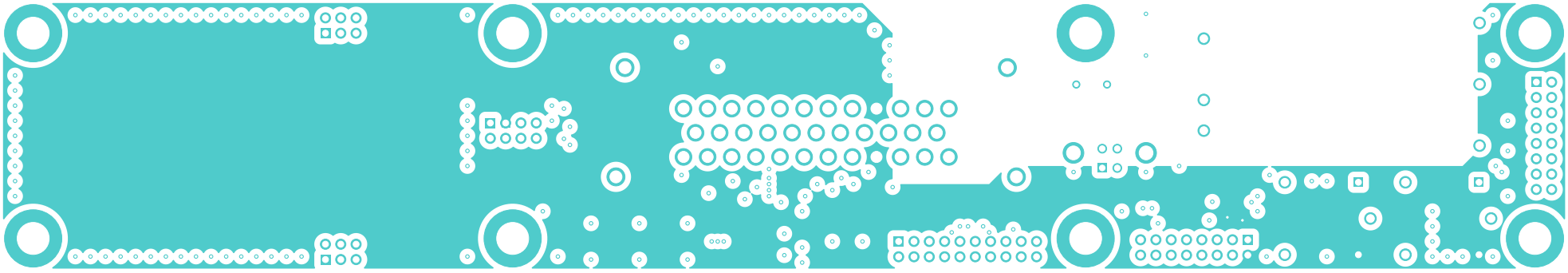


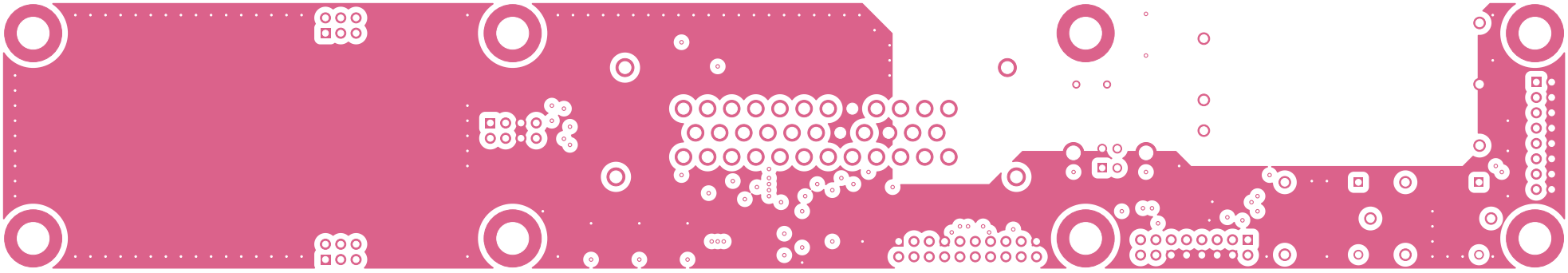


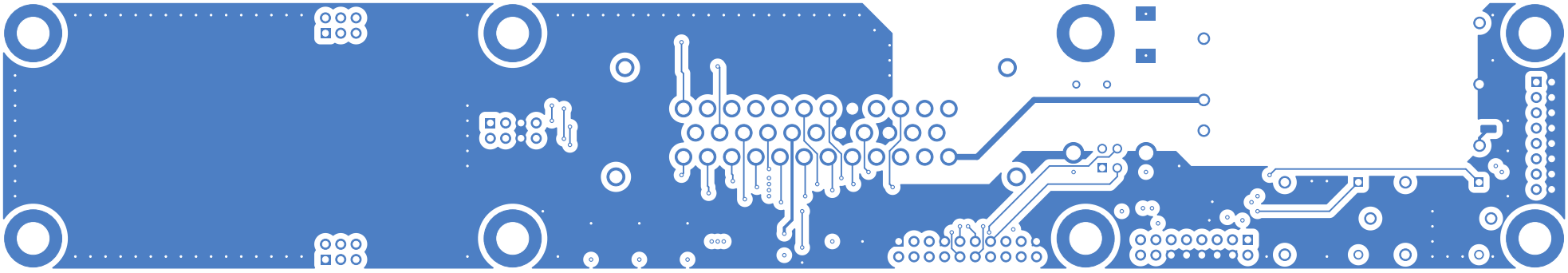


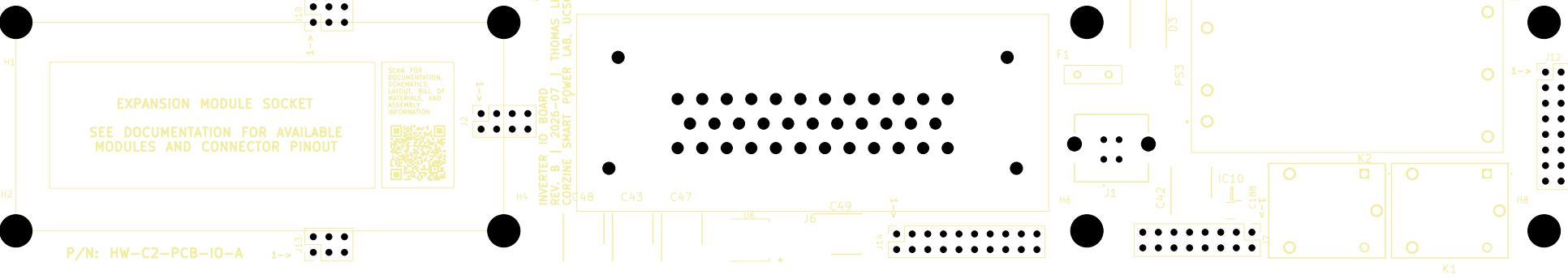












INVERTER I/O BOARD
REV. B | 2026-07 | THOMAS LIAC
CORZINE SMART POWER LAB, UCSC

EXPANSION MODULE SOCKET
SEE DOCUMENTATION FOR AVAILABLE
MODULES AND CONNECTOR PINOUT



P/N: HW-C2-PCB-IO-A

H1

H2

H3

H4

H5

F1

PS3

H7

H8

H6

J1

K2

K1

IC10

C188

J12

J14

C48

C43

C47

J6

C49

C42

J10

1->

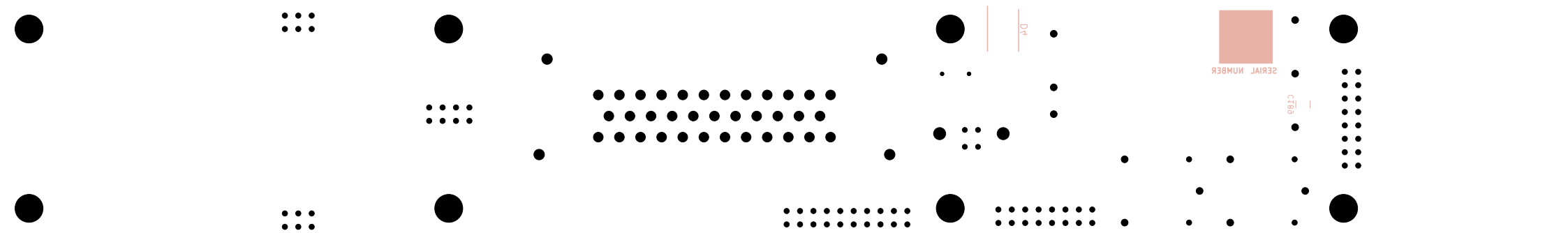
J2

1->

J13

1->

D3

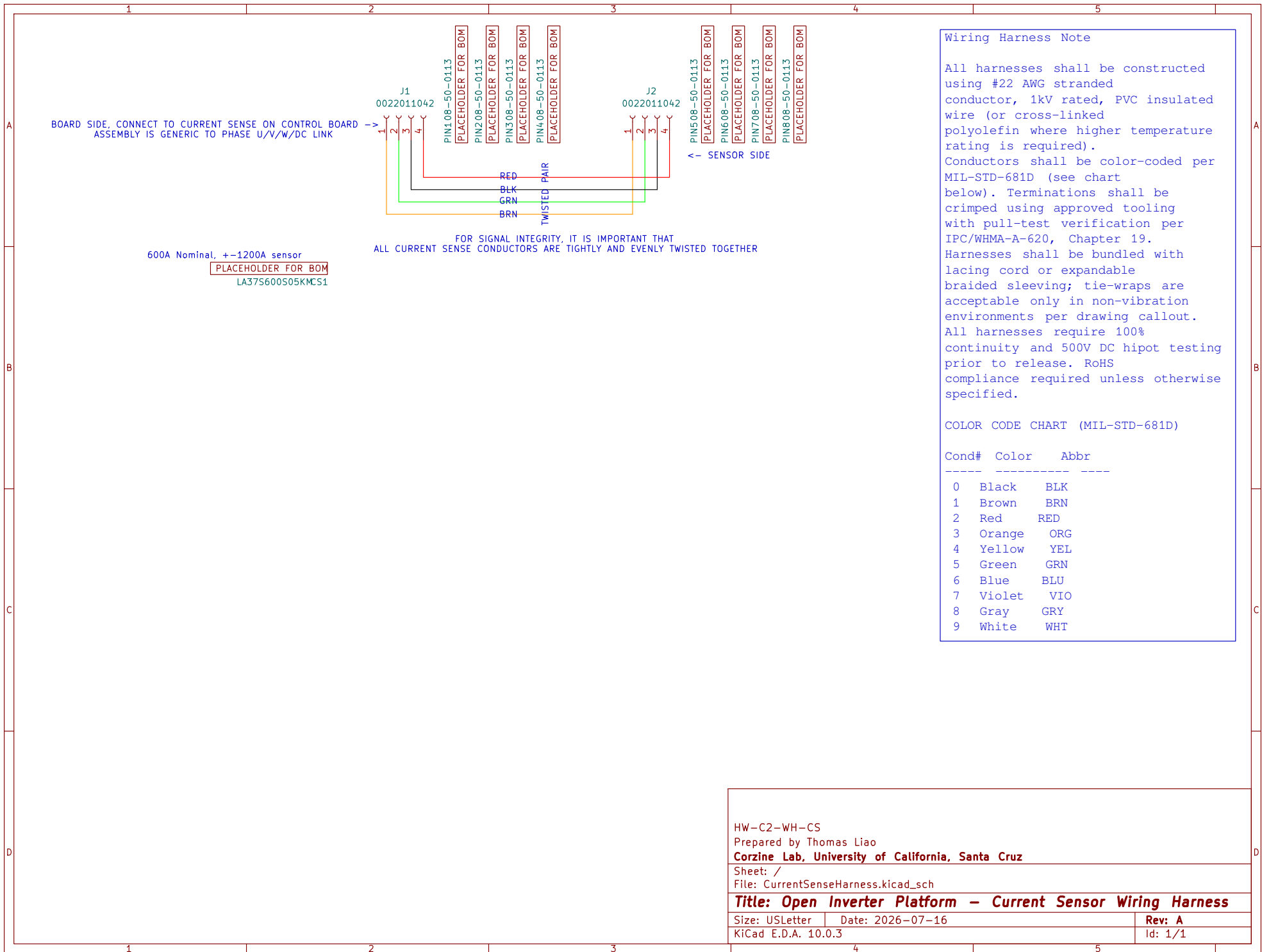


CAPACITORTEMPSENSEHARNESS

HW-C2-WH-TSBCM-A · wiring harness — schematic

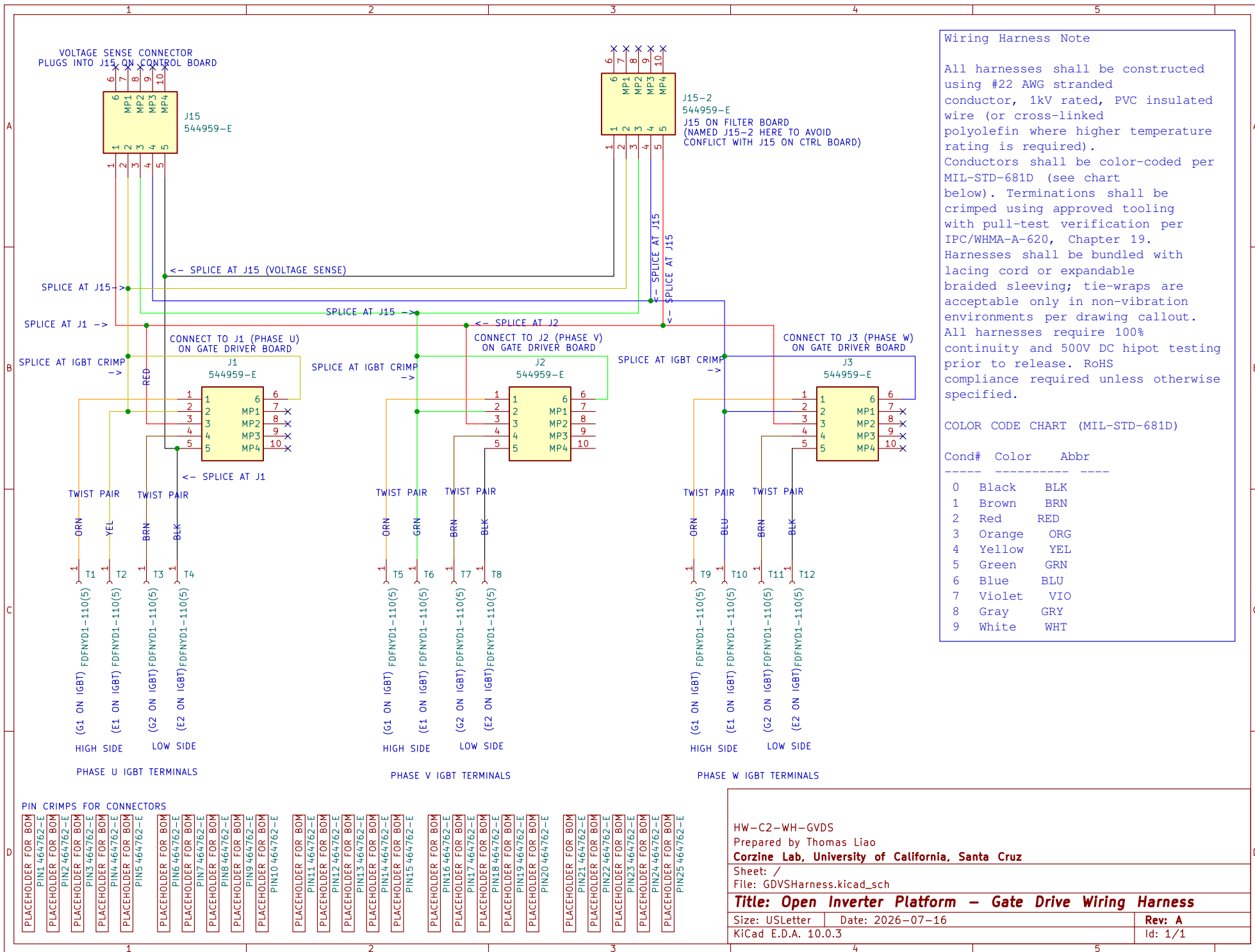
CURRENTSENSEHARNESS

HW-C2-WH-CS-A · wiring harness — schematic



GDVSHARNESS

HW-C2-WH-GDVS-A · wiring harness — schematic



TEMPSENSEHARNESS

HW-C2-WH-TSIGBT-A · wiring harness — schematic

